AxoSim: Learned Neuron Models for Population Simulation at Scale

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Abstract

Detailed neuron simulators resolve morphology and ion-channel mechanisms, but repeated numerical integration is costly at population scale. Learned surrogates reduce this cost; in recurrent circuits, however, each prediction also affects the model's future inputs. AxoSim combines morphology-conditioned GRU and Mamba neuron models with delayed synaptic routing, mutable efficacies, and differentiable population state. AxoSim–GRU Medium reaches a spike Mean F1 of 0.61 compared with up to 0.45 for released Branch-ELM and TCN models on their respective own training datasets, while processing 16.8 million neuron-steps per second in the matched inference benchmark, more than 20 times Branch-ELM's throughput. On an RTX 5090, the runtime simulates 65,536 neurons with 4,000 synaptic contacts per neuron at 3.38× real-time speed, whereas a million-neuron population can be simulated in real time for a low number of synaptic contacts. On the same consumer-grade GPU, populations of 10,000 neurons are trained in real time with full backpropagation through time (BPTT). In a temporal-shift experiment on Spiking Heidelberg Digits (SHD), morphology and synaptic adaptation raise target accuracy by 6.46 ± 1.20 percentage points across six independently trained models. A separate corpus provides 50,000 validated population traces from 100 recurrent connectomes with connectome-disjoint splits. Together, these experiments separate teacher-forced fidelity from recurrent simulation and inference-time adaptation.

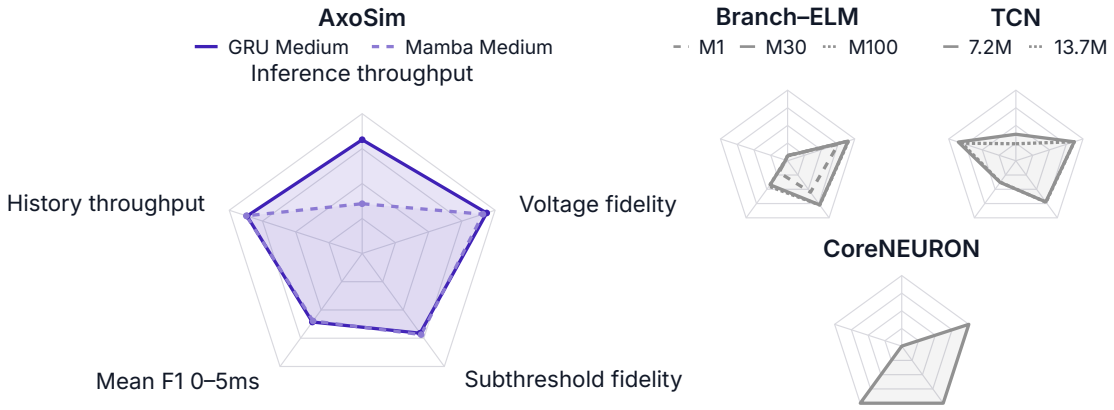


Figure 1: The axes show (i) inference throughput, counting 1-ms neuron predictions per second; (ii) voltage fidelity, the voltage SERA skill as a fraction of the skill of the teacher voltage clipped below action potentials; (iii) subthreshold fidelity, the RMSE skill of the voltage outside action-potential windows; (iv) spike F1 averaged over 0–5 ms matching tolerances; and (v) history throughput, counting input milliseconds processed per second for a 2-second history. The fidelity axes are dataset-relative: each learned model is scored on held-out

traces of its own training data, AxoBench for AxoSim and NeuronIO for Branch–ELM and TCN, with firing rates matched between the datasets (Section 6.2). CoreNEURON is the ground-truth simulator; Branch–ELM, a learned surrogate of this simulator and a predecessor of AxoSim, and TCN are released reference models. AxoSim–GRU Medium and Mamba Medium are compared with three Branch–ELM sizes and the smallest and largest TCN (7.2M and 13.7M parameters); larger radii indicate better performance. Appendix A.14 contains the raw measurements, the runtime contracts and the mapping from measurements to plotted values.

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1 Overview

1.1 The simulation problem

Biophysically detailed neuron simulators integrate membrane mechanisms over a morphology, which preserves a direct relation between model parameters and cellular processes. That detail is indispensable when the question concerns a named conductance, compartment, or synaptic mechanism, but it also makes repeated simulation and gradient-based fitting expensive as the number of cells grow. Learned neuron models amortize simulation by approximating the map from synaptic input histories to somatic voltage and spikes [3, 21].

Most evaluations of learned neurons compare the predicted and teacher responses on fixed input traces, measuring response quality of a single neuron. A recurrent population introduces an additional difficulty because the behavior at population level is not inferable in a straightforward way. A predicted spike is delayed, delivered to downstream dendrites, and incorporated into the input for later predictions. An approximation error or bias can therefore starkly alter or trivialize population dynamics. Modeling a population of neurons, therefore, necessitates additional evaluation in an appropriately different way and invites population-aware training.

AxoSim is a neuron simulation platform incorporating this distinction and providing capabilities for both settings. It complements the mechanistic teacher with compatible interfaces for fitting neuron responses, connecting the fitted models through explicit synaptic contacts, advancing persistent populations, and differentiating through a trajectory. CoreNEURON simulations of Hay layer-5 pyramidal cells provide the principal training targets, while the released Branch-ELM (expressive leaky memory) models are used as architectural starting point and baseline.

1.2 From single neurons to recurrent populations

At the single-neuron level, each AxoSim model receives excitatory or inhibiory events at a dendritic destination, combines them through a morphology-conditioned front end, and predicts somatic voltage and spike logit at every millisecond. The models have been designned to prioritize response fidelity by default, and the AxoSim–Lite specialization with reduced state and computation is optimized for population inference, but preserves these event semantics.

The population runtime maintains recurrent neuron state and a directed graph of synaptic contacts. Each contact is described by its source, sign, branch destination, delay, and efficacy. During each simulation step, spike events occur when each neuron's spike prediction exceeds a fitted threshold. The runtime routes those events after their delays, accumulates their efficacies at the target branches. For large-population inference, fused GPU kernels organize connected execution in a phase- and tile-major layout.

The bulk of parameters affecting response behavior are shared for a reduced memory footprint. Notably, there are dedicated parameter groups for a neuron's morphology, for adaptation and individual synaptic contacts. The latter two groups exist primarily for adaptation at inference-time or to model heterogeneous behavior between neurons, but all are amendable to gradient-based optimization, in the population setting too. This separation allows the same interface to express morphology transfer, changes in synaptic efficacy, and task adaptation while preserving their interpretation as predictive coordinates.

1.3 Evidence across fidelity, execution, and adaptation

AxoBench introduced spike Mean F1 across 0–5 ms matching tolerances, Voltage SERA, and Dynamics SERA as the core metric set for single-neuron evaluation [27]. In Section 6, these three quantities are plotted on efficiency axes (parameter count, inference-time computation and throughput). The largest separation from Branch-ELM occurs in spike timing. Voltage SERA also decreases substantially, Dynamics-SERA less notably.

Paired synaptic interventions extend this analysis to changes in the teacher's response. Population timing then covers a complete recurrent step at explicit neuron and contact counts, while connected diagnostics measure activity after predictions enter the feedback loop. Adaptation analyses test transfer to excluded morphologies, a controlled conductance shift and a temporal distribution shift on Spiking Heidelberg Digits dataseet. Table 1 associates each result with its measurement condition.

Table 1: Principal AxoSim contibutions.

Area	Result	Measurement condition
Neuron-modeling fidelity	Mean F1 0.522; the lowest AxoSim Voltage SERA is 21.3% below its Branch-ELM reference, and the lowest Dynamics SERA is 4.4% below its reference.	Ordinary AxoBench, held-out morphology-balanced traces.
Intervention susceptibility	Response SERA is 9.57–10.70% below the strongest Branch-ELM response reference, with lower error in all twenty paired model comparisons.	350 scoring pairs across five input interventions and five morphologies; 50 separate baseline calibration pairs.
Population inference at scale	1,048,576 neurons advance at 1.177 billion neuron-steps/s with 362 incoming synaptic contacts per neuron; 65,536 neurons with 4,000 contacts advance at 3.38 times real time.	Complete RTX 5090 step with delayed routing, mutable efficacies, state, and 1% telemetry.
Population data	50,000 detailed-teacher traces from 100 recurrent connectomes assigned to disjoint context splits.	Five excitatory morphologies and one inhibitory morphology; complete connectomes define the split.
Adaptation	SHD target accuracy improves by 6.46 ± 1.20 percentage points; 20,000 neurons with 2,000 synaptic contacts complete a 256-step full-BPTT update in real time.	Six independently trained SHD source models; complete full-BPTT update on one RTX 5090.
Biological transfer	A small 58-parameter adapter improves spike and dynamics fidelity on five excluded morphologies; synaptic parameters recover one controlled conductance shift.	Predictive transfer is established; .

2 System design

2.1 Architecture

The two model families proposed (AxoSim–GRU and AxoSim–Mamba) share a dendritic front end, which converts synaptic events into a temporal feature sequence, and differ in the temporal core that integrates this sequence; separate linear heads decode the resulting features into somatic voltage and spike logit (Figure 2). The front end extends the branch organization of Branch-ELM, in which each routed synapse is assigned to one model branch and the weighted inputs of a branch are summed before temporal integration [21]. AxoSim adds two components to this organization, namely morphology-conditioned input gains and a learned saturation of each branch sum, followed by a projection of saturated branch values to the temporal feature width. The main departure from Branch-ELM therefore lies in how anatomy enters the model: one set of weights serves several morphologies, and each morphology contributes a description of its input mapping in addition to its input events. The ablations in Section 2.3 identify this anatomical conditioning as the most consequential addition, and Appendix A.3 gives the dimensions and pretraining of the four reported models.

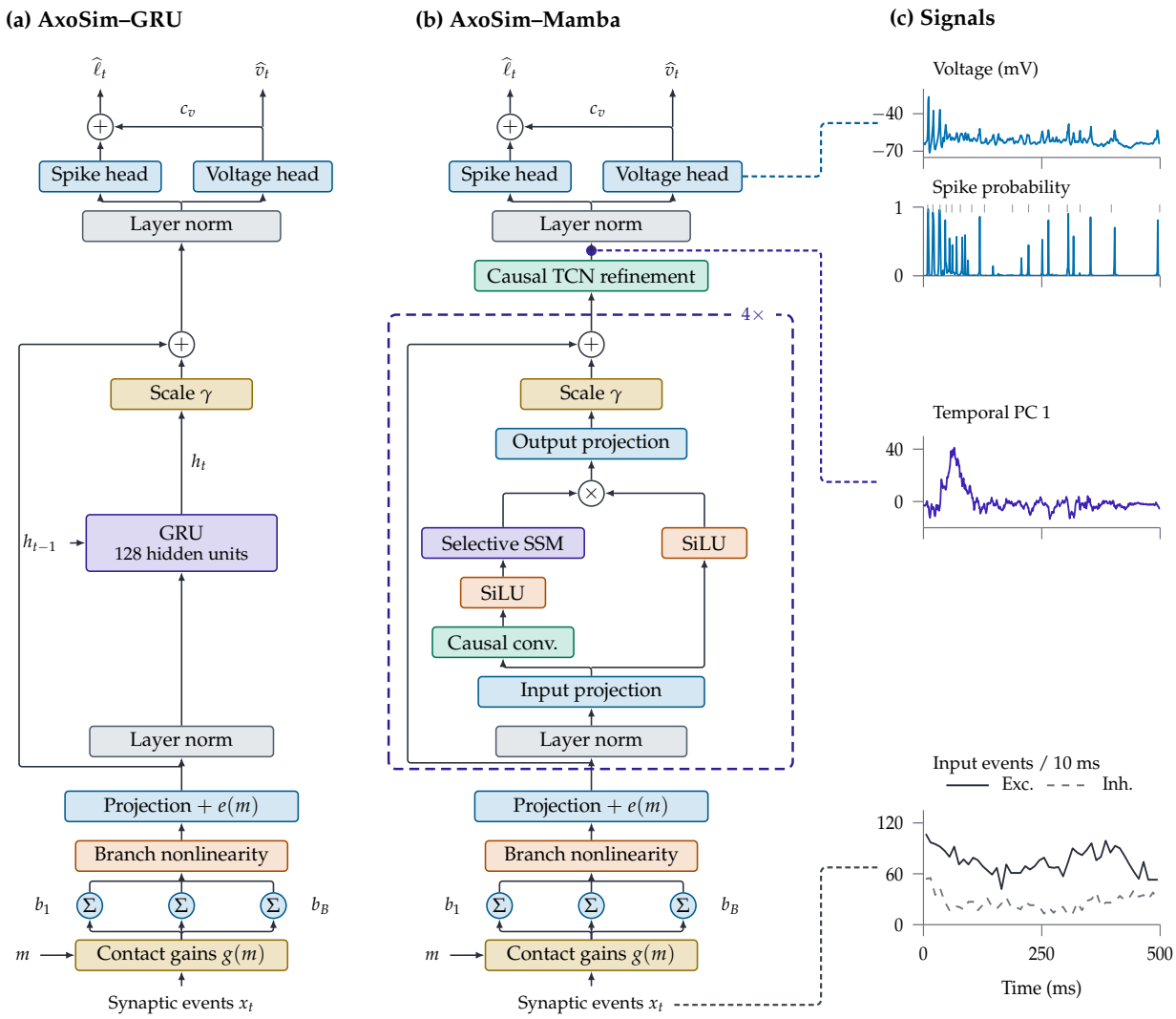


Figure 2: Neuron model architectures and a measured response. (a,b) Morphology-conditioned branch aggregation precedes residual temporal transformations and separate output heads. Branch assignment and aggregation follow Branch-ELM. Mamba Medium has four temporal blocks followed by causal temporal refinement (Table 11); γ is a learned channel-wise residual scale, and c_v is the voltage-to-logit coupling coefficient. **(c)** Dashed lines associate the Mamba stages with one held-out 500-ms trace (Appendix A.13). Outputs and the first principal component of temporal features are shown at 1-ms resolution; input events use 10-ms counts. Gray ticks mark detailed-simulator spikes.

Because one model serves several morphologies, the same input site reaches different dendritic segments in different cells, and its somatic effect differs accordingly. To represent this dependence, each synaptic input is multiplied by two positive gains before branch aggregation. The first is learned separately for every morphology and synaptic slot. The second is computed from fixed descriptors of the anatomical input mapping, which comprise route occupancy, the input sign, the dendritic region and the mean length of the mapped segments. Route occupancy records whether any mapped dendritic segment uses an input site and how many segments share it. A two-layer perceptron with 16 hidden units, shared across morphologies and input sites, maps these descriptors to a log-gain, and its exponential gives the second gain. Note that both gains are positive by construction, such that a

morphology can change the sensitivity of an input site but cannot reverse the excitatory or inhibitory type of its events.

Each gain-weighted branch sum then passes through the saturating function, a hyperbolic tangent with learned scale and offset, which bounds the branch response before linear mixing across branches. The mixed values are projected to the temporal feature width D, and a morphology embedding is added to the projection. This embedding is a learned D-dimensional vector selected by the morphology index of the neuron; it is constant over time and shared by all neurons of that morphology, so it shifts their temporal inputs by a common morphology-specific offset.

2.2 Sequence modeling cores and readout heads

The two temporal cores follow their standard formulations. The GRU core combines its retained state with a candidate update through reset and update gates [5], whereas the Mamba core applies state-space transitions whose parameters depend on the current input [9]. During streaming inference, both cores advance their state one input step at a time. When the complete input history is available in advance, as for a recorded trace, the Mamba recurrence can instead be evaluated in parallel over time with a selective scan. Figure 1 reports the resulting history throughput as the number of input milliseconds processed per second and more details are in Appendix A.14. Because each state update is affine in the previous state, the scan merely regroups the composition of these updates, so a Mamba model can be pretrained with a parallel scan to increase pretraining speed, but can still be advanced step-by-step at inference time.

Each temporal core is placed in a pre-normalized residual block. For the sequence z_ℓ of D-dimensional representations at depth ℓ, the block computes

$$z_{\ell+1} = z_\ell + \gamma_\ell \odot \mathcal{M}_\ell(\mathrm{LN}(z_\ell)),$$

where LN denotes layer normalization, $\mathcal{M}_\ell$ the GRU or Mamba transformation, and $\gamma_\ell \in \mathbb{R}^D$ a learned vector that scales each output channel, with the elementwise product $\odot$ broadcast across time. The identity term in the derivative of this block gives gradients a direct path across depth [11]; gradients across time, in contrast, still pass through products of recurrent transition derivatives [16].

In AxoSim–Mamba, three causal convolution blocks with dilations 1, 4, and 16 follow the Mamba blocks. Each block combines a width-three depthwise convolution with a pointwise channel projection, and together the three blocks span 43 samples. Conceptually, these blocks mix the most recent features at several local time scales, complementing the longer history carried by the recurrent state.

At time index t, separate linear heads with weights w_v, w_s and biases b_v, b_s decode the final normalized feature vector $r_t \in \mathbb{R}^D$ into voltage and spike logit,

$$\begin{aligned} \widehat{v}_t &= w_v^\top r_t + b_v, \\ \ell_t^{\mathrm{head}} &= w_s^\top r_t + b_s, \\ \widehat{\ell}_t &= \ell_t^{\mathrm{head}} + c_v \widehat{v}_t. \end{aligned}$$

Here $\widehat{v}_t$ is the predicted voltage in scaled units, which correspond to $10\widehat{v}_t - 67.7\,\mathrm{mV}$, ℓ_t^{head} is the logit of the spike head alone, and $\widehat{\ell}_t$ is the final spike logit. The coupling coefficient c_v is fixed at 4 following empirical parameter search. Because the final spike logit depends on the predicted voltage, the spike objective also trains the voltage head, while the spike head contributes a learned term that is added to the voltage-dependent part of the logit. The coupling thus ties spike prediction to the predicted membrane potential, and removing it degrades both the voltage and the spike scores (Section 2.3).

2.3 Component ablations

To measure the contribution of individual components, a two-layer Mamba model is retrained with one component disabled at a time, while the allocated parameter count and the number of pretraining presentations are held fixed. Figure 3 reports the resulting changes in Voltage Root-SERA and in the normalized trapezoidal average of spike F1 over 0–5 ms. Both quantities differ from the metrics used in the main evaluation, and Appendix A.6 gives the pretraining and scoring protocol together with their relation to Voltage SERA and arithmetic Mean F1.

Removing route-occupancy conditioning causes the largest penalties among the five components, increasing Voltage Root-SERA by 0.423 mV and reducing trapezoidal F1 by 0.087. Removing the morphology embedding, local temporal refinement, branch saturation or spike–voltage coupling also degrades both quantities, and in all five comparisons the paired 95% intervals of both penalties exclude zero. Notably, the most consequential component adds no temporal capacity; it supplies fixed information about which dendritic segments each input reaches. The morphology embedding, which also encodes cell identity, causes the second-largest voltage penalty.

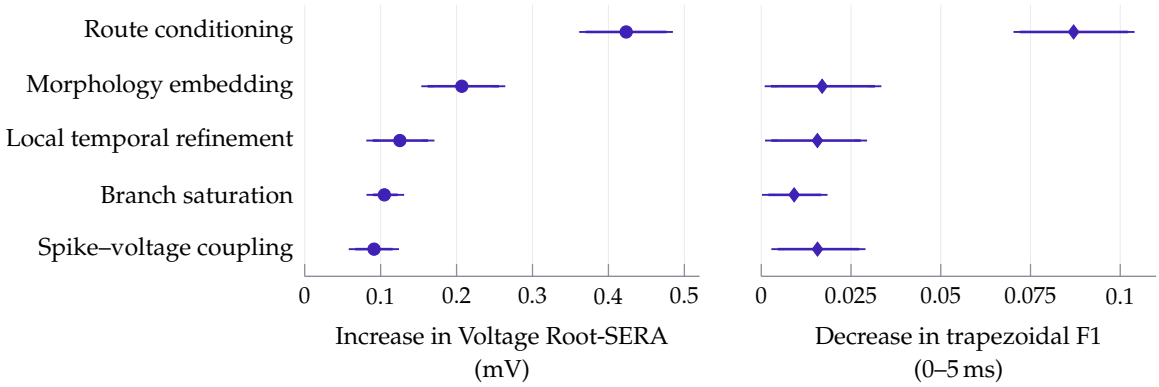


Figure 3: Component-removal penalties after matched retraining of a two-layer Mamba model. Voltage Root-SERA penalties are ablated minus full and trapezoidal-F1 penalties are full minus ablated, so positive values favor the retained component. Points give estimates and whiskers paired 95% trace-bootstrap confidence intervals for one common pretraining seed.

3 The population runtime

The models of Section 2 primarily discussed the models proposed for the single-neuron simulation setting. These update a deep temporal state at every millisecond, which suits single-neuron fidelity but limits the size of a simulated population and its simulation speed. AxoSim–Lite is the complementary population model: it keeps eight recurrent values per neuron (Appendices A.7 and A.4) and updates them once every four milliseconds. This section describes how the runtime schedules these updates, routes the resulting spikes and stores connectivity. Two design choices make large populations feasible. First, the voltage signals are processed, predicted and routed in patches. The forecast horizon (patch size) matches the minimum synaptic delay, so the forecasts of different neurons within one patch are mutually independent. Second, contact topology is computed on demand, so per-contact memory reduces to a single efficacy.

3.1 Patchified processing and delayed feedback

Let x_t denote the branch inputs of a neuron at native millisecond t and $\widehat{y}_t$ its predicted spike logit and voltage. One Lite update integrates a completed four-millisecond patch of branch inputs and

predicts the outputs for each millisecond of the following patch. If τ is the patch boundary and h_n the persistent neuron state, the update computes

$$(\widehat{y}_\tau, \ldots, \widehat{y}_{\tau+3}, h_{n+1}) = F_\theta(x_{\tau-4:\tau-1}, h_n).$$

Each of the four forecasts keeps its own millisecond, so a spike predicted at $\tau + 2$ is emitted at $\tau + 2$ and the patch structure does not coarsen spike timing.

The forecast horizon P is chosen not to exceed the minimum synaptic delay $d_{\min}$ of the network, here both 4 ms, so that a spike predicted anywhere within the current patch reaches its target only after that patch has ended (Figure 4 and Appendix A.9). Consequently, no forecast depends on a spike predicted within the same patch, and all neurons can compute their forecasts independently and in parallel. Parallel simulators of detailed networks rely on the same condition to integrate neurons independently between spike exchanges [12]; AxoSim satisfies it with a learned model that forecasts an entire patch in one update.

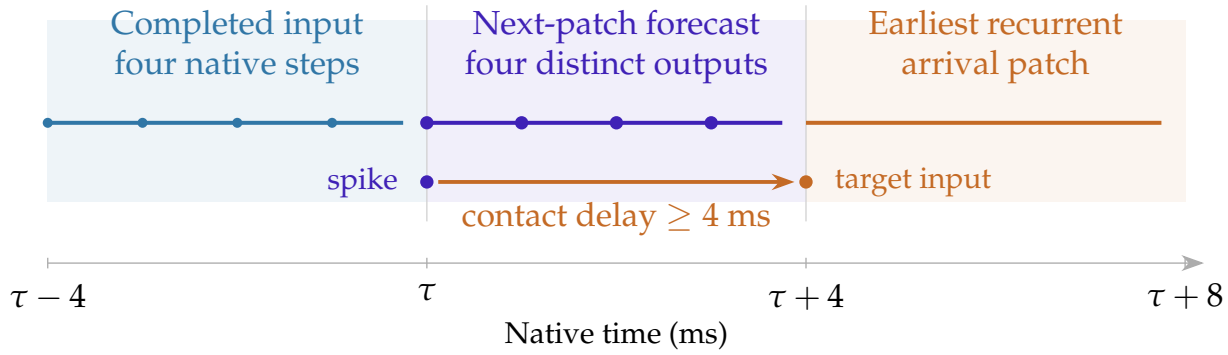


Figure 4: Causal scheduling of a four-step neuron forecast. At boundary τ, the completed input patch supplies four distinct predictions. Even the earliest predicted spike, at τ, cannot reach another neuron before $\tau + 4$ under the minimum 4-ms contact delay. Later predicted spikes arrive correspondingly later.

A connected simulation step combines these forecasts with spike routing. Each neuron is assigned a phase between 0 and 3, which determines the millisecond within each four-millisecond cycle at which it performs its patch update. Neuron indices are partitioned into spatial tiles, contiguous blocks of 4,096 neurons that also structure the connectivity described below. Phases are balanced within every tile and within each morphology and excitatory or inhibitory group, so every millisecond updates about one quarter of the population with an unchanged composition, and the recurrent work is spread evenly over time. Neurons are stored in memory ordered by phase and tile, so that the neurons updated in one step occupy a contiguous range, which the fused GPU kernels process together. A forecast spike logit above the fitted threshold of its neuron defines a spike event. The runtime then adds the signed, efficacy-weighted contribution of each event to the target branch in a delay ring, a circular buffer of branch inputs per neuron that is indexed by arrival time. Finally, voltage samples of a selected subset of neurons are copied asynchronously to host memory. The population timings in Section 8 include all of these operations, so they measure complete simulation steps.

3.2 Procedural contacts with mutable efficacies

At several thousand incoming synaptic contacts per neuron, explicit source and target index arrays would occupy 8 bytes per contact with two 32-bit indices, eight times the 8-bit efficacy itself. In the structured population graphs used for the scaling measurements, a deterministic arithmetic rule therefore replaces these arrays. From the neuron indices, the excitatory or inhibitory role of the source

and an offset into the destination input channels, the forward rule computes the source, sign, destination branch and delay of each contact, and its inverse enumerates the targets of an emitted event. In this framework, tiles reflect the spatial neighborhood of neurons within a population: In the million-neuron benchmark (Appendix A.17), 90% of a neuron's outgoing contact channels target neurons in its own tile with delays of 4 or 5 ms, and the remaining channels target another tile with delays of 6 or 7 ms. Every contact nevertheless keeps its own efficacy, which the delay-ring accumulation reads together with the computed contact identity. This separation permits per-contact adaptation (Section 4), since efficacies can be changed or trained while the topology remains implicit. Reconstructed connectomes, which follow no such rule, still require explicit contact indices.

Importantly, because the topology is computed rather than stored, each contact costs only its efficacy and the routing of its events, so the number of synaptic contacts per neuron enters two tradeoffs, one with memory and one with speed. Figure 10 shows both: populations trade neurons against contacts, from 1,048,576 neurons with 362 contacts to 65,536 with 4,000, and throughput falls with connectivity, as does training speed (Section 9). With K incoming contacts, the deployed 8-bit efficacy bank requires K bytes per neuron plus one shared scale. Recurrent state and the delay ring instead depend on state width, branch count and ring length. For 4,000 contacts, the efficacies occupy 4,000 bytes per neuron, the delay ring 1,888 bytes and the eight-value BF16 recurrent state 16 bytes. At this density the efficacies make up two thirds of the per-neuron memory, whereas at the 362 contacts of the million-neuron benchmark the delay ring dominates. Appendix A.8 lists the remaining coefficient storage and the gradient and optimizer storage required during adaptation. Contact density thus determines most of the per-neuron memory and the routing work, while the shared neuron-model weights are stored once for the entire population.

4 Heterogenity and adaptation

Biological neurons are (very) heterogeneous, but making all parameters amendable to change would make simulation of large populations intractable on a single GPU. For this reason, the AxoSim runtime shares the bulk of parameters across neurons. Dedicaed small parameter groups are not shared and target differences in learned behavior, structural classes, or changes in synaptic efficacy. Broadly, they are fitted in the same way as the pretraining of the unshared groups, via gradient-based optimization.

4.1 Neuron, structural-class and contact parameters

The adaptive coefficients act on four stages of the Lite update, namely input sensitivity, memory decay, the recurrent proposal and the output readout. Let u_n be the input token, i.e., the four-millisecond patch of branch inputs projected to width D, and let h_n be the persistent state. Token offsets and gains transform u_n into $\widetilde{u}_n$, and a decay shift δ_d, a proposal gain g_h and a proposal offset o_h modify the state transition,

$$h_{n+1} = d \odot h_n + (1 - d) \odot \left[(1 + g_h) \odot \tanh(W_u \widetilde{u}_n + W_h h_n + b_h + o_h)\right], \qquad d = \sigma(\lambda + \delta_d).$$

Here W_u, W_h and b_h are the shared projections and bias, λ is the shared decay logit, and σ is the logistic sigmoid, so δ_d shifts the memory timescale of each state coordinate. Output gains and offsets on the readout complete seven coefficient groups. For token width $D = 24$, state width $H = 8$, forecast horizon $P = 4$ and two outputs per native step, these groups contain $2D + 3H + 2P \cdot 2 = 88$ coefficients.

The population interface assigns these coefficients to the three levels at which neurons of a population differ, namely individual cells, structural classes and synapses (Table 2). A behavior bank holds one

coefficient vector per neuron and a morphology bank one vector per structural class, and the coefficients of neuron i are the sum of its behavior row and the row of its class. A synaptic bank assigns a log-efficacy $s_{i,k}$ to contact k of neuron i, and the positive efficacy $\exp(s_{i,k})$ multiplies the incoming event before branch integration. The levels differ strongly in size: at 4,000 contacts per neuron, the synaptic bank holds 45 times as many parameters as the behavior bank, so synaptic adaptation inherits the contact tradeoffs of Section 3. The neuron and class banks, in turn, are not separately identifiable, since a vector added to one class row and subtracted from all of its neuron rows leaves every response unchanged. Attributing a fitted change to a class or to individual neurons therefore requires an additional constraint (Appendix A.7).

Table 2: Adaptive parameter ownership in the compact population interface. The seven neuron/class coefficient groups are detailed in Appendix A.7.

Group	Ownership	Coefficients	Effect
Behavior	one neuron	88 per neuron	input, memory and readout
Morphology	one structural class	88 per class	class-shared response
Synaptic	one contact	one per contact	positive input efficacy

4.2 Full temporal gradients and captured updates

Gradients are computed for a supplied event history: the input events of every neuron are given, as in a recorded or teacher-generated trajectory, and full backpropagation through time runs from the sequence loss through every recurrent state transition and branch aggregation to the contact efficacies. Connected inference, in contrast, forms discrete events by thresholding the predicted logits, and the gradient does not pass through this thresholding step. Training therefore optimizes the response to given input histories and does not differentiate through spikes that the population generates itself.

Because full backpropagation through time retains the activations of every time step, memory is quickly the limiting resource. A naive implementation would form a dense neuron–time–contact matrix, but accumulating sparse event indices and values at the branches of each neuron before temporal integration partially reduces immediate memory pressure and was adopted.

An adaptive update consists of many short GPU kernels, so kernel-launch overhead can dominate its runtime. The accelerated path therefore captures the complete update, from input processing, forward execution and loss to backward propagation, gradient clipping and the optimizer step, in one CUDA Graph, using PyTorch's fixed-shape, fixed-address replay mechanism [18].

Capture accelerates the update without changing its result. In a matched 40-update SHD comparison (Appendix A.10), eager and captured execution end with identical parameters. Replay takes 1.84 s against 6.40 s for eager execution, after a separate capture preparation of 0.74 s. Section 9 reports real-time full-BPTT measurements with dense contacts.

5 Evaluation methodology

5.1 Separating fitting, selection, and evaluation

Each comparison separates data used to fit parameters, choose a model or adaptation schedule, and estimate final performance. The unit of separation follows the scientific question. Ordinary AxoBench uses morphology-balanced traces, population specialization uses recurrent contexts, SHD uses disjoint examples and independently trained source models, and connected evaluation uses unseen connectomes. Appendix A.1 records the exact partitions.

For the ordinary comparison, development data fix the model identity and threshold-calibration rule. The first 24 final traces determine one prevalence-matched threshold, and the remaining 96 provide all reported fidelity metrics. Released Branch-ELM models and the four released TCN checkpoints of Beniaguev et al. [3] are evaluated unchanged. Both baselines were trained on the original NeuronIO cell, which is not among the five evaluation morphologies. The TCNs receive binary inputs, and their somatic output is read as the membrane potential relative to $-67.7\,\mathrm{mV}$, as in their training code. On the original NeuronIO test traces, this interface reproduces the published spike AUC of 0.99 for the TCN and 0.95 for Branch-ELM M1 (Appendix A.16). Table 3 compares the training recipes of the three model families.

Population-input specialization selects one learning-rate schedule per architecture family on development contexts and applies it unchanged on the final contexts. SHD selects the adaptive groups, learning rate, and update count on one source model, then reuses that choice for five independently trained sources. Connected-population comparisons pair alternatives on each unseen connectome. Final data are therefore used only after model and schedule choices have been fixed.

5.2 Neuron and population fidelity

The single-neuron evaluation follows the AxoBench core metric set [27]. Spike Mean F1 averages event F1 over integer matching tolerances from 0 through 5 ms,

$$\mathrm{MeanF1}_{0:5} = \frac{1}{6}\sum_{\tau=0}^{5} F_1(\tau).$$

Predicted and teacher events are matched one-to-one in temporal order within $\pm\tau$. The six constituent values distinguish exact event recovery from agreement that appears only at a permissive tolerance.

For prediction error $e_i = \widehat{y}_i - y_i$ and teacher-derived relevance $r_i \in [0,1]$, the conditional squared error above threshold u is

$$E(u) = \frac{1}{|\{i : r_i \geq u\}|}\sum_{i:r_i\geq u} e_i^2.$$

SERA is the support-normalized integral of $E(u)$, evaluated at 101 evenly spaced relevance thresholds. Voltage SERA applies the construction to somatic voltage and has units of mV^2. Dynamics SERA applies it to adjacent native-time voltage changes and has units of $\mathrm{mV}^2/\mathrm{ms}^2$. SERA already names the squared quantity, and Root-SERA denotes its square root. Lower SERA is better.

The subthreshold error measures the voltage between action potentials, which SERA weights little. It is the RMSE of predicted and teacher voltage, both clipped at $-55\,\mathrm{mV}$, over the steps outside teacher and predicted action-potential windows. Each window starts 2 ms before a spike and ends, per cell type, where the teacher's mean spike waveform first falls below $1\,\mathrm{mV/ms}$ in slope. The subthreshold skill $1 - \mathrm{RMSE}/\mathrm{RMSE}_0$ compares it with the error of predicting each trace's mean clipped voltage. Higher skill is better.

Response SERA evaluates paired changes in voltage after an intervention. It compares the predicted intervention-minus-baseline change with the teacher's change, using relevance derived from both teacher trajectories and their difference. This tests the accuracy of the response to altered input; its units are mV^2.

Connected evaluation keeps three discrepancies separate. A morphology term compares morphology-conditioned firing rates with fixed targets, a temporal term compares input-count autocorrelations at lags from 1 to 256 ms, and a stationarity term measures full-to-tail drift in global and role-conditional activity. Global rate, input load, variability, and active fraction are also checked for collapse or composition changes. The equations and fixed normalizations appear in Appendix A.20.

5.3 Uncertainty and runtime contracts

Single-neuron differences use 1,000 paired bootstrap resamples of the 96 scored traces, stratified by morphology. All three metrics are recomputed from each paired sample. These intervals describe trace-level uncertainty for fixed pretrained models. Intervention comparisons instead resample 175 paired shard clusters within 25 morphology–condition strata and use simultaneous intervals for all twenty AxoSim–Branch-ELM response differences. Retraining variation requires independently fitted models, as used for adaptation, where the report gives the mean and sample standard deviation across source models. Population-training results use the paired outcome on every unseen connectome. The reported uncertainty therefore follows the source of variation in each analysis.

Runtime quantities are attached to explicit workloads. Isolated-neuron rates use float32 and matched batch and sequence shapes. Population inference uses the p90 latency of a complete step, including event discovery, delayed routing, branch accumulation, mutable efficacy, recurrent state, and 1% telemetry. Population adaptation measures a complete 256-step update, including transfer, sparse input construction, forward execution, loss, full-sequence backward propagation, clipping, and optimization. Hardware, precision, repeats, and exclusions are tabulated in Appendix A.2.

CoreNEURON, Branch-ELM, and AxoSim use different workloads and hardware, so Figure 1 compares their directly measured quantities. A cross-system speed ratio is reserved for a future shared-workload measurement. Raw values and axis mappings appear in Appendix A.14.

Table 3: Training recipes of the compared models. Compute is written as epochs $\times$ cells $\times$ presentations per cell and epoch $\times$ window length, where an epoch is one pass over the usable 5.5 s of every training trace; AxoSim and Branch-ELM sample windows at random positions, so their epochs count passes in expectation. Branch-ELM and TCN values follow their released training configurations [3].

	AxoSim	Branch-ELM	TCN
Models	GRU Small, GRU Medium, Mamba Small, Mamba Medium	M1–M100	four released checkpoints
Training data	AxoBench v1 train: 5 cells, 100,000 traces	NeuronIO train: 1 cell, 37,120 traces	NeuronIO train: 1 cell, 38,400 traces
Compute	29.1 (GRU Small) or 36.4 epochs $\times$ 5 cells $\times$ 220k $\times$ 0.5 s	6.9 epochs $\times$ 1 cell $\times$ 408k $\times$ 0.5 s	0.38 epochs $\times$ 1 cell $\times$ 528k $\times$ 0.4 s
Optimizer	AdamW	Adam	Nadam
Batch size	32	8	8
Learning rate	6×10^{-4}, cosine decay	5×10^{-4}, cosine decay	10^{-4} stepped to 10^{-6}
Regularization	weight decay 10^{-4}	none	L2 penalty 10^{-8} per layer
Losses	voltage MSE, spike BCE, SERA and F1 surrogates (Appendix A.3)	spike BCE, voltage MSE	spike, voltage and dendritic-voltage losses

6 Single-neuron fidelity and efficiency

6.1 Held-out fidelity and the practical model frontier

The ordinary AxoBench comparison measures fidelity on five Allen morphologies on which AxoSim was pretrained and which the released Branch-ELM and TCN models did not see; Section 6.2 reports each family on the other's data. Appendix A.3 describes the four AxoSim models and their pretraining, and Appendix A.1 the trace partition. In this comparison, the largest quantitative separation

occurs in spike timing (Table 4). AxoSim–GRU Medium reaches Mean F1 0.522, whereas the strongest released Branch-ELM spike reference reaches 0.077, an absolute difference of 0.445. At the same time, GRU Medium lowers Voltage SERA from 307.18 to 241.61 mV^2, a 21.3% reduction. Mamba Small lowers Dynamics SERA from 448.57 to 428.72 mV^2/ms^2, a smaller 4.4% reduction. The dominant improvement is therefore event recovery, accompanied by a substantial voltage-trajectory gain and a modest improvement in native-time voltage changes.

Table 4: Held-out ordinary-AxoBench results. Lower SERA and higher Mean F1 and throughput are better. Inference is measured in millions of neuron-steps/s.

Model	Parameters	Voltage SERA	Dynamics SERA	Mean F1	Inference
AxoSim–GRU Small	17,432	243.92	429.12	0.486	**21.79**
AxoSim–GRU Medium	122,744	**241.61**	429.13	**0.522**	16.81
AxoSim–Mamba Small	32,536	244.29	**428.72**	0.454	18.86
AxoSim–Mamba Medium	1,128,200	244.46	439.00	0.520	4.40
Branch-ELM M30	10,952	307.41	448.57	0.077	0.742
Branch-ELM M50	19,252	307.18	448.58	0.076	0.735
TCN 7_128_153	9,234,582	309.31	448.55	0.078	2.96
TCN 7_292_169	13,717,058	309.35	448.56	0.085	2.75
TCN 8_224_217	7,219,778	311.70	448.59	0.077	4.71
TCN 9_256_241	9,416,770	315.13	448.57	0.078	3.82

For AxoSim–Mamba Medium, Mean F1 is 0.443 higher than the strongest Branch-ELM spike reference, with a paired morphology-stratified 95% bootstrap interval of [0.422, 0.463] (Section 5). Its Voltage-SERA difference from the strongest released voltage reference is $-62.72\,mV^2$, with a 95% interval of $[-69.06, -56.72]$. AxoSim–GRU Medium has the highest final Mean-F1 point estimate, 0.0016 above Mamba Medium. The near tie in spike fidelity is practically important because the GRU model is much smaller and nearly four times faster at inference.

The released TCNs place the original deep-network surrogate of this teacher on the same axes. With 7.2–13.7 million parameters and 7.2–13.7 billion forward-pass FLOPs per 500-step window (Appendix A.15), they exceed GRU Medium by a factor of about 60 to 115 in both quantities, yet their fidelity on the five evaluation morphologies matches Branch-ELM: Voltage SERA is 309.31–315.13 mV^2 and Mean F1 is 0.077–0.085. Since both baselines were fitted to a single cell, whereas AxoSim was pretrained on the evaluation morphologies, this comparison reflects the pretraining distribution as much as the architecture. It shows that a hundredfold larger surrogate does not compensate for the missing morphologies.

The target quantity determines model choice (Figure 5). GRU Medium leads Mean F1 and Voltage SERA, GRU Small leads inference throughput, and Mamba Small records the lowest Dynamics SERA. Mamba Medium provides a state-space alternative within 0.002 of the spike leader, at 4.40 million neuron-steps/s. The three leaders define the measured frontier. The larger Mamba state offers temporal capacity at a higher execution cost.

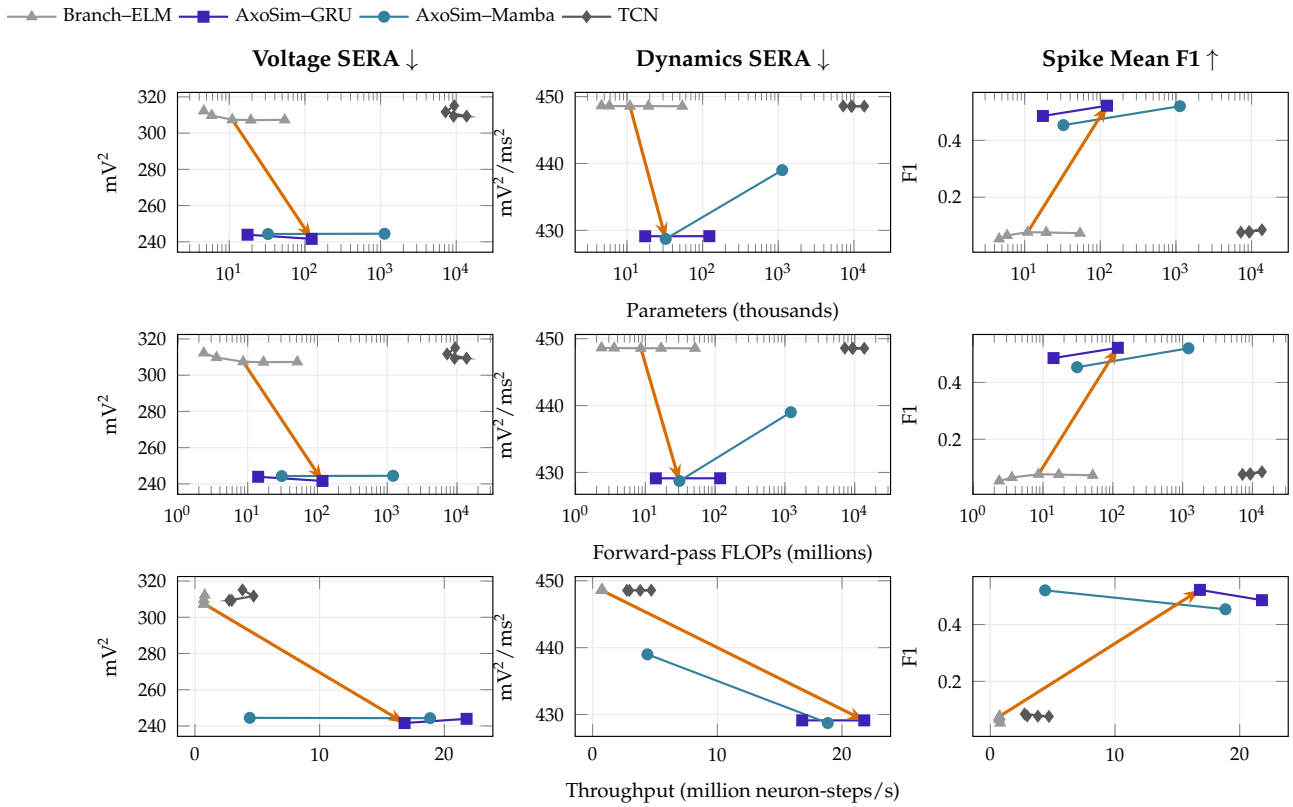


Figure 5: Accuracy–cost tradeoffs for AxoSim and released Branch–ELM and TCN models. Voltage SERA, Dynamics SERA and spike Mean F1 are shown against trainable parameters, forward-pass FLOPs and inference throughput. Lines connect models from the same architecture family, and orange arrows link the Pareto-selected Branch–ELM and AxoSim models. FLOPs are the operations recognized by `torch.profiler`, plus one-dimensional convolutions, for a batch-one, 500-step forward pass with 1,278 input channels. Throughput uses an RTX 5090, batch 32, 500 steps and float32. The Pareto rule appears in Appendix A.15.

6.2 Fidelity depends on the evaluation cells

Learned neuron models are usually evaluated on held-out traces of the cell they were trained on, so their reported fidelity also reflects how far the evaluation data lie from the training data. Scoring each family on the other's data makes this dependence visible (Table 5). Because SERA values depend on the voltage statistics of each dataset, the table reports the voltage skill $1 - \mathrm{SERA}/\mathrm{SERA}_0$, where SERA_0 belongs to a constant-voltage prediction on the same traces (Appendix A.16).

Table 5: Voltage skill and Mean F1 on the 1,536 NeuronIO test traces of the original training cell and on the AxoBench confirmation block of five morphologies. AxoSim ranges span its five morphology identities on NeuronIO.

	NeuronIO cell		AxoBench morphologies	
Model	Voltage skill	Mean F1	Voltage skill	Mean F1
Branch-ELM M50	0.498	0.286	0.138	0.076
TCN 7_292_169	0.470	0.232	0.132	0.085
AxoSim–GRU Medium	0.266–0.357	0.015–0.146	0.322	0.522
AxoSim–Mamba Medium	0.231–0.339	0.023–0.137	0.314	0.520

Each family is most accurate on the cells it was trained on. On its NeuronIO cell, Branch-ELM M50 removes 50% of the constant-prediction voltage error and reaches Mean F1 0.286; on the AxoBench morphologies, these values fall to 14% and 0.076. The skill of Branch-ELM on its own data thus exceeds the 32% that GRU Medium reaches on AxoBench by a factor of about 1.5, although the comparison favors Branch-ELM, since NeuronIO contains a single cell without synaptic delay and is the simpler prediction problem. Spike timing shows the opposite picture. Because the constant prediction cannot place spikes in time, its Mean F1 is 0.002 on NeuronIO and 0.007 on AxoBench, so the spike skill $(\mathrm{F1} - \mathrm{F1}_0)/(1 - \mathrm{F1}_0)$ nearly equals Mean F1. GRU Medium reaches a spike skill of 0.518 on its AxoBench morphologies, 1.8 times the 0.284–0.286 that Branch-ELM M50 and M100 reach on their own cell and 2.2 times the 0.231 of the strongest TCN. The spike advantage therefore persists under the comparison that favors the references, whereas the voltage advantage does not. Conversely, AxoSim's spike fidelity drops to 0.015–0.146 on the NeuronIO cell, whereas its voltage skill of 0.23–0.36 remains comparable to its AxoBench value. The NeuronIO labels mark threshold crossings, whereas the AxoBench labels mark action-potential peaks, and NeuronIO synapses act without the 1-ms delay of AxoBench, so part of this spike-timing loss reflects these conventions rather than the cell. Fidelity of a learned neuron is therefore a property of the model together with its evaluation cells, and the transfer comparison of Table 4 measures, in addition to architecture, how well pretraining on several morphologies covers new ones.

The raw skills above still mix the model with its dataset. Three dataset differences change the scores independently of the model, and the dataset-relative comparison removes each of them (Appendix A.16). First, the teacher voltage clipped below action potentials, the best prediction that emits no spike, already removes 53% of the constant-prediction voltage error on AxoBench and 68% on NeuronIO, so voltage skill is expressed as a fraction of this ceiling. Second, AxoBench traces fire more often; only traces below 6 Hz, which both datasets cover, are scored, and NeuronIO traces are reweighted to AxoBench's firing-rate distribution. Third, the subthreshold error excludes action-potential windows that end, per cell type, where the teacher's mean spike waveform flattens. Under these adjustments, the GRU models reach 0.936 and 0.937 of the voltage ceiling against 0.909 for Branch-ELM M100, with non-overlapping 95% intervals, and Mamba Medium matches M100 at 0.908 (Table 6). AxoSim's spike Mean F1 of 0.56–0.61 exceeds Branch-ELM's highest value of 0.45 and the TCNs' 0.37–0.38. The subthreshold skill of 0.68–0.72 lies below Branch-ELM M30–M100 at 0.77–0.78 and within the TCN range of 0.67–0.73. Figure 1 plots these dataset-relative scores.

Table 6: Dataset-relative fidelity on each model's own training data, with 95% bootstrap intervals. Voltage is the SERA skill as a fraction of the clipped-teacher skill; subthreshold is the RMSE skill outside action-potential windows. All scores use traces below 6 Hz, reweighted to AxoBench's firing-rate distribution.

Model	Data	Voltage	Subthreshold	Mean F1
AxoSim–GRU Small	AxoBench	0.936 [0.917, 0.955]	0.690 [0.681, 0.699]	0.572 [0.543, 0.598]
AxoSim–GRU Medium	AxoBench	0.937 [0.919, 0.956]	0.705 [0.695, 0.715]	0.607 [0.581, 0.633]
AxoSim–Mamba Small	AxoBench	0.935 [0.917, 0.952]	0.683 [0.675, 0.690]	0.564 [0.537, 0.590]
AxoSim–Mamba Medium	AxoBench	0.908 [0.881, 0.933]	0.718 [0.709, 0.727]	0.598 [0.574, 0.620]
Branch-ELM M1	NeuronIO	0.753 [0.746, 0.760]	0.552 [0.545, 0.560]	0.206 [0.198, 0.214]
Branch-ELM M10	NeuronIO	0.877 [0.873, 0.881]	0.739 [0.734, 0.745]	0.312 [0.302, 0.323]
Branch-ELM M30	NeuronIO	0.901 [0.898, 0.904]	0.773 [0.767, 0.779]	0.421 [0.410, 0.433]
Branch-ELM M50	NeuronIO	0.908 [0.904, 0.911]	0.779 [0.773, 0.785]	0.444 [0.431, 0.458]
Branch-ELM M100	NeuronIO	0.909 [0.906, 0.912]	0.776 [0.769, 0.782]	0.447 [0.436, 0.459]
TCN 7_128_153	NeuronIO	0.878 [0.874, 0.882]	0.714 [0.705, 0.722]	0.374 [0.365, 0.384]
TCN 7_292_169	NeuronIO	0.871 [0.867, 0.875]	0.727 [0.721, 0.733]	0.375 [0.366, 0.386]
TCN 8_224_217	NeuronIO	0.869 [0.865, 0.873]	0.721 [0.716, 0.727]	0.379 [0.369, 0.389]
TCN 9_256_241	NeuronIO	0.852 [0.847, 0.856]	0.669 [0.664, 0.674]	0.379 [0.369, 0.389]

6.3 Where the fidelity gains arise

Mean F1 averages six spike-matching tolerances, and AxoSim's advantage appears at every setting. GRU Medium exceeds Branch-ELM M30 at every tolerance in Figure 6. At exact timing, F1 is 0.261 versus 0.047, and at 5 ms it is 0.641 versus 0.102. The throughput-leading GRU Small has the same qualitative separation from the closest-size Branch-ELM reference.

Conditional squared-error curves provide the complementary voltage result. Integration of each conditional squared-error curve yields SERA. Across most of the relevance support, AxoSim has lower conditional voltage error than Branch-ELM, so the Voltage-SERA reduction is distributed across a broad threshold range. The smaller separation between the dynamics curves agrees with the modest aggregate Dynamics-SERA improvement. Thus, event recovery can improve markedly even when error in native-time voltage changes decreases only slightly.

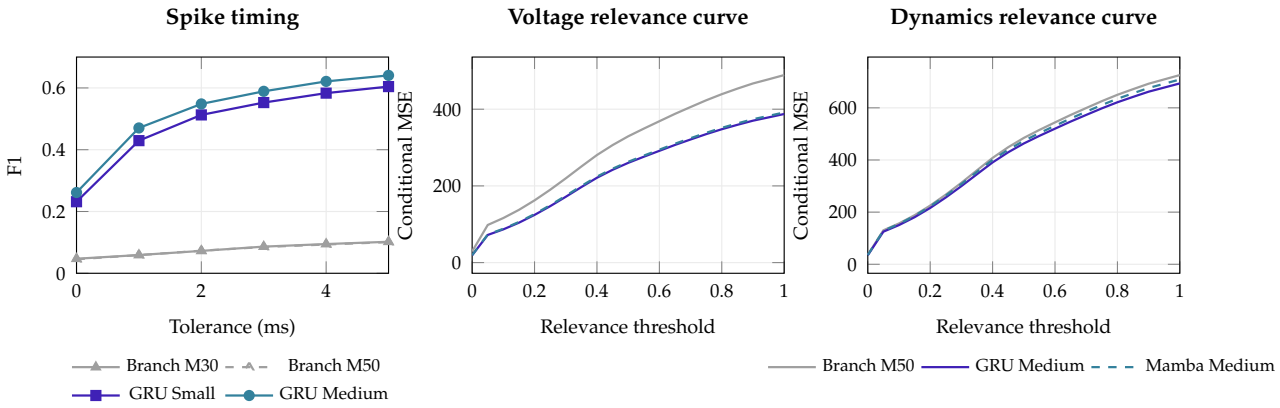


Figure 6: Constituents of the core fidelity metrics. Spike F1 is reported across matching tolerances, followed by the conditional squared-error curves integrated by Voltage and Dynamics SERA.

7 Intervention-response fidelity

7.1 Predicting the teacher's response to changed synaptic input

Controlled changes in synaptic input provide a direct test of response prediction. For each baseline–intervention pair, Response SERA compares the predicted voltage change with the detailed teacher's change using the AxoBench relevance weighting [27]. Four AxoSim models and five released Branch-ELM sizes are evaluated on the same 350 paired histories spanning event removal, excitatory removal, inhibitory removal, site silencing and temporal jitter across five morphologies. Model weights remain fixed, and 50 separate baseline pairs determine the spike thresholds before scoring (Appendix A.11).

AxoSim reaches Response SERA 255.55–258.78 mV^2, compared with 286.17 mV^2 for the strongest Branch-ELM response model. The resulting reductions are 9.57–10.70%, and all twenty AxoSim–Branch-ELM paired comparisons have simultaneous 95% intervals below zero. A zero-voltage-change prediction gives 298.27 mV^2, providing a reference for the amount of response captured. Spike Mean F1 on the intervened traces is 0.416–0.491 for AxoSim, versus 0.062–0.076 for Branch-ELM.

The condition-wise firing changes make these responses concrete (Figure 7). Removing inhibitory events increases the teacher's mean firing rate by 4.36 Hz; GRU Medium and Mamba Medium predict increases of 3.09 and 3.45 Hz, while Branch-ELM M1 predicts 17.98 Hz. Conversely, removing excitatory events decreases teacher activity by 3.47 Hz, compared with 3.36 and 3.47 Hz for the two AxoSim models. Event removal and site silencing likewise produce teacher-aligned decreases. These observations relate the output-fidelity comparison to the expected effects of changing excitation and inhibition.

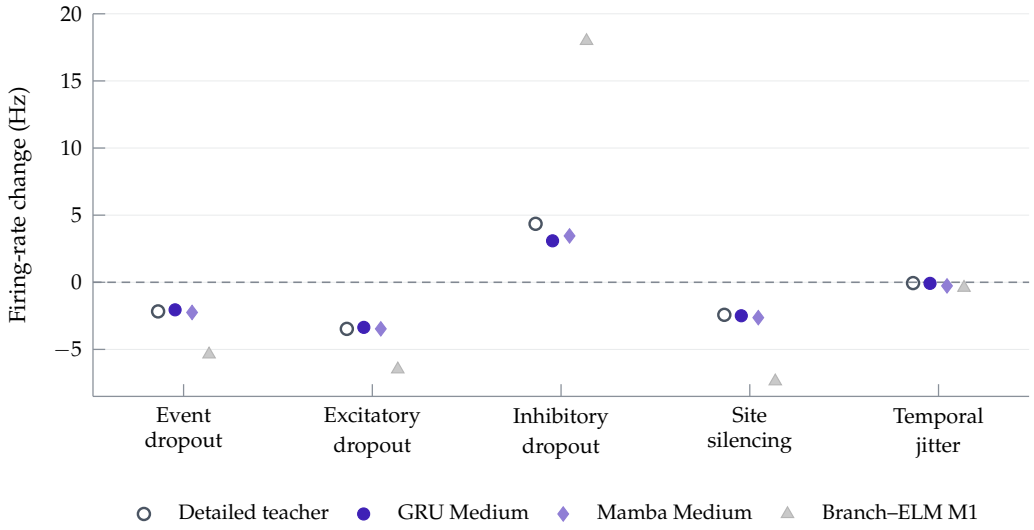


Figure 7: Mean firing-rate changes under five synaptic-input interventions. Points show the detailed teacher and three models on the same 350 paired scoring histories, excluding the first 500 ms of each six-second history. Thresholds are fixed from separate baseline calibration pairs. Branch-ELM M1 has the lowest pooled Response SERA among the released Branch-ELM models. Full response metrics and sampling are given in Appendix A.11.

Predicted voltage changes also correlate more closely with the teacher, with centered Pearson correlations of 0.483–0.487 for AxoSim versus 0.331–0.336 for Branch-ELM. At the same time, the least-squares response gain through zero remains below one, at 0.269–0.303 for AxoSim and 0.179–0.186 for

Branch-ELM, where a gain of one denotes the teacher's voltage-change scale. Correlation and gain therefore describe complementary aspects of the lower Response SERA.

7.2 Internal-state diagnostics of intervention responses

Mamba Medium's internal states provide a complementary description of the predicted responses. A separate analysis (Appendix A.13) uses all 12,500 intervention pairs to form 1,575 disjoint cohort signatures, each averaging the paired temporal state displacement of seven or eight traces over successive 500-ms windows. The three-dimensional UMAP projection shows intervention-dependent regions while marker shape identifies morphology (Figure 8a; 15). Quantitative comparisons use the original representation space.

Temporal trajectories use the same cohort signatures, weighted by their seven or eight constituent pairs. The first three principal components explain 93.4% of the variance of the standardized cohort-averaged state displacements, and condition means trace the evolution of the response at 500-ms intervals (Figure 8b). Gray arrows summarize average next-window displacement within occupied spatial bins, following low-dimensional analyses of recurrent dynamics [6, 23]. Across the intervention–morphology groups, within-intervention directions exceed between-intervention directions by 0.563 in cosine similarity, describing consistent response organization across morphologies.

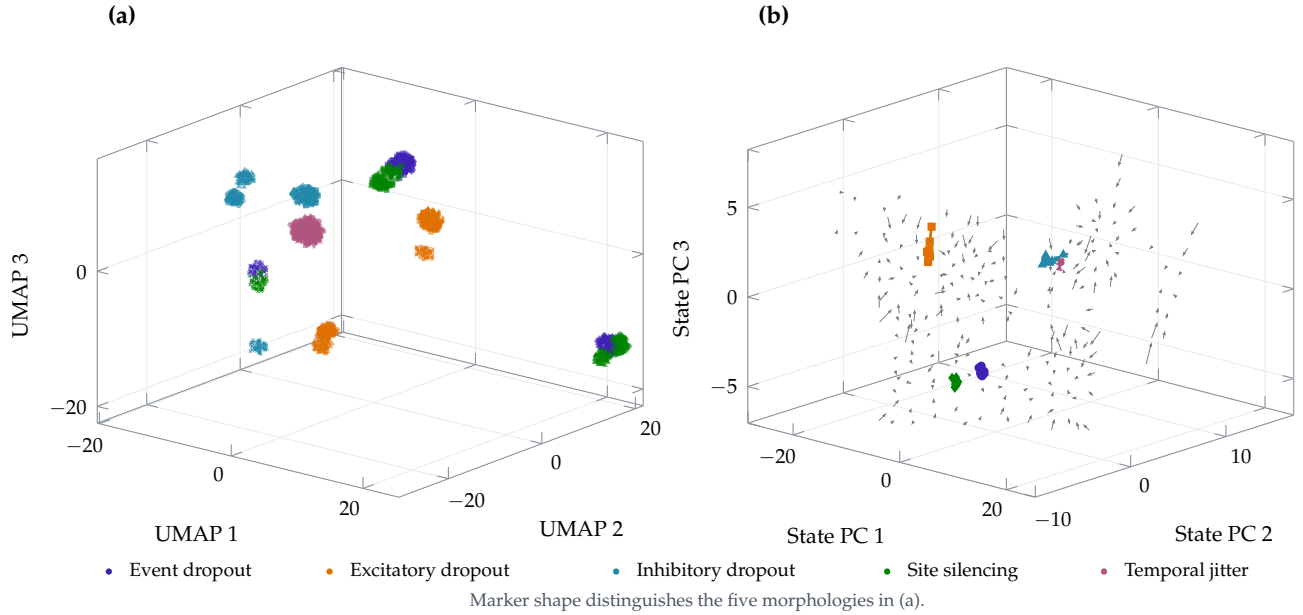


Figure 8: State-response diagnostics for Mamba Medium. (a) Three-dimensional UMAP of 1,575 disjoint cohort-averaged signatures from 12,500 pairs. Color denotes intervention and marker shape denotes morphology. (b) Trace-count-weighted cohort responses in three PCA coordinates, with 202 gray arrows showing measured mean movement in occupied spatial bins. The first three components explain 93.4% of the variance of cohort-averaged state displacements. Output-response metrics are reported separately in Appendix A.11.

An external response distance links this organization to teacher outputs, following scientific-model manifold comparisons [17] and representational similarity analysis [13]. Across all 300 pairs of intervention–morphology groups, original-space temporal-state geodesics correlate 0.446 with the teacher's voltage-and-spike response distance. The corresponding branch representation reaches 0.211, and the input-perturbation descriptor reaches 0.471. Temporal-state distances therefore as-

sociate more closely with teacher responses than branch distances, while input descriptors show a comparable association. The distance construction, paired uncertainty and all three comparisons appear in Appendix A.13.

7.3 Response susceptibility across intervention strength

The comparisons above apply each intervention at a single strength. A model can match the teacher at that strength while scaling its response incorrectly, so dose series measure how the predicted response grows with the size of the perturbation. For excitatory removal, inhibitory removal and site silencing, the 70 released scoring pairs of each condition in `axobench-interventions` are re-simulated with the detailed teacher at additional strengths, namely 10, 20, 50, 75 and 100% of the events of one sign removed and 10, 50 and 75% of the sites silenced. Each series retains the released baseline history and intervention seed, so the removed events are nested across strengths and the released strength reproduces the released pair exactly. Models, spike thresholds and response statistics are those of the matched comparison, and the released TCN [3] is selected by the rule that selected Branch-ELM M1 (Appendix A.12).

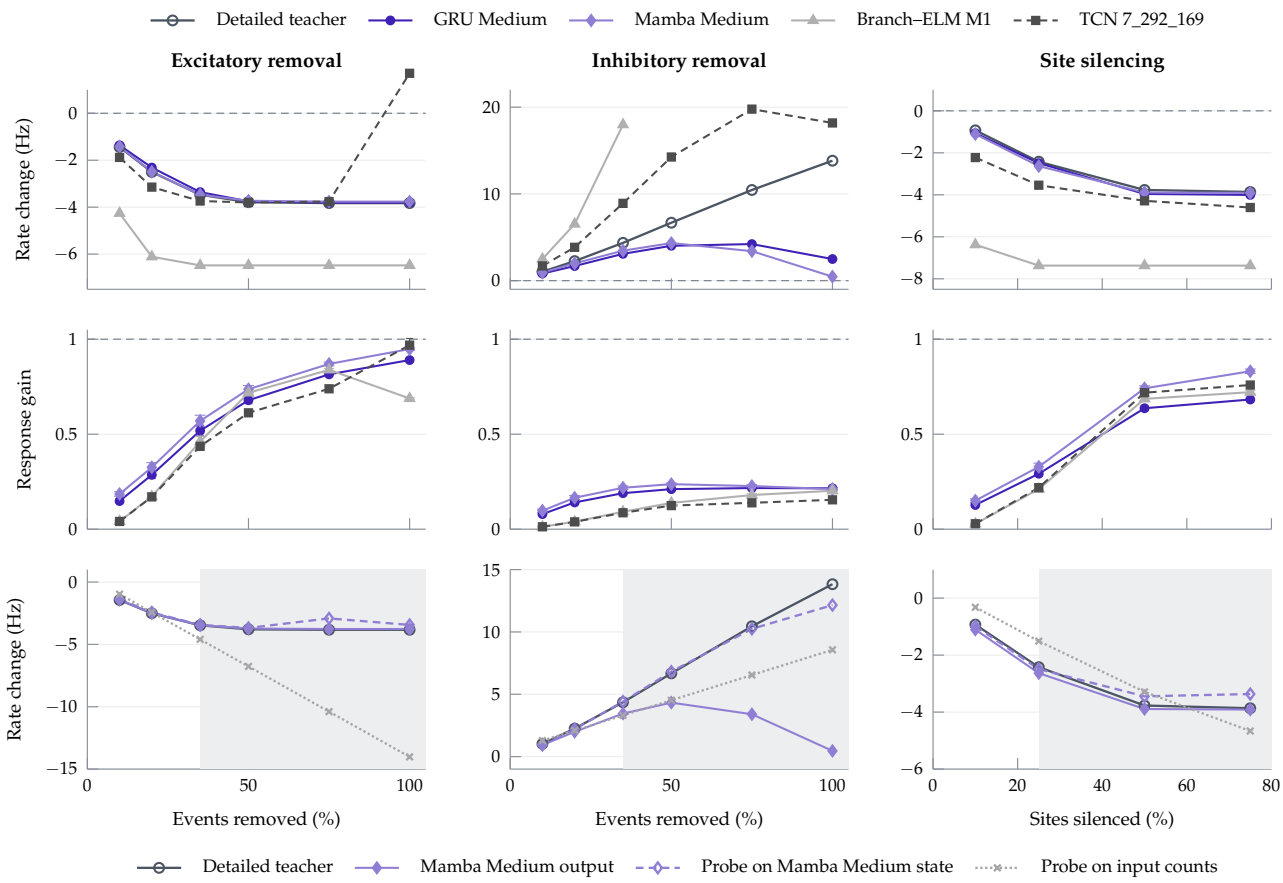


Figure 9: Responses to three input interventions as a function of strength. Each point pools the same 70 scoring pairs per condition, and the released strengths are 35% removal and 25% silencing. **Top:** decoded firing-rate change; Branch-ELM M1 leaves the inhibitory panel above 35% removal, at 43, 160 and 410 Hz. **Middle:** least-squares response gain of the voltage change, where one denotes the teacher's scale; whiskers are 95% intervals from 1,000 shard-cluster bootstrap draws within morphologies. **Bottom:** the teacher's firing-rate change, Mamba Medium's own prediction, and two linear probes fitted to the teacher's rate change on the unshaded strengths of all three conditions and evaluated on the shaded ones. One probe reads Mamba Medium's time-averaged temporal state, and the control reads the change in excitatory and inhibitory input counts.

When an intervention reduces synaptic drive, AxoSim reproduces the teacher's firing response over the whole range of strengths (Figure 9). Under excitatory removal, the teacher's rate falls by 1.44 Hz at 10% and by 3.83 Hz once the cell stops firing; GRU Medium and Mamba Medium predict decreases of 1.36 and 1.42 Hz at 10% and of 3.81 and 3.76 Hz at full removal. Site silencing yields the same agreement. Branch-ELM M1 predicts decreases of up to 6.48 and 7.37 Hz in the two conditions, because its baseline rate exceeds the teacher's by a factor of 1.7 to 1.9. The selected TCN follows the teacher up to 75% excitatory removal but predicts a rate increase of 1.70 Hz at full removal.

The response gain locates the voltage attenuation reported in Section 7 at small perturbations. Under excitatory removal, the gain of the four AxoSim models rises from 0.13–0.19 at 10% to 0.86–0.95 at full removal, and under site silencing from 0.11–0.15 at 10% to 0.68–0.83 at 75%. Weak perturbations are therefore reproduced at a fraction of their size, whereas large reductions of drive are reproduced almost in full. Branch-ELM M1 and the TCN start lower, with gains of 0.03–0.04 at 10%, and reach comparable values only at strong perturbations.

Removing inhibition separates the regime in which AxoSim is valid from the regime in which it is not. The teacher's rate increases steadily, from 1.03 Hz at 10% removal to 13.83 Hz at full removal. AxoSim follows this increase up to the released strength, predicting 3.09–3.66 Hz against 4.36 Hz at 35%, but its prediction peaks at 4.0–4.8 Hz at 50% and falls to -0.23 to 2.49 Hz at full removal. Branch-ELM M1 and the TCN overshoot instead, with Branch-ELM M1 reaching 410 Hz. The boundary is set by the teacher's output regime and not by the novelty of the input. Measured against 12,500 ordinary histories from the training generator, the mean inhibitory event rate falls below that of every ordinary history at 35% removal, and excitatory removal leaves the ordinary range of the excitation-to-inhibition ratio from 20% onward, yet AxoSim remains accurate in both cases. The failure begins instead where the teacher's absolute firing rate leaves the ordinary range: 10.4 Hz at 50% removal lies at the 98.8th percentile of ordinary rates, 14.2 Hz at 75% lies at the 99.99th percentile, and 17.6 Hz at full removal exceeds the largest ordinary rate of 14.5 Hz.

The voltage and spike outputs fail differently in this regime. Under full inhibitory removal, AxoSim predicts a mean depolarization of the clipped somatic voltage of 4.21–4.85 mV, larger than the teacher's 2.83 mV, while its predicted firing collapses. The models therefore predict a sustained depolarization without the corresponding spikes, and in the drive-reducing conditions their mean voltage change stays within 1.7 mV of the teacher's.

Mamba Medium's temporal state retains the information that its spike output loses. The RMS displacement of the channel-standardized state increases monotonically with strength in all three conditions, from 0.61 at 10% to 2.56 at full inhibitory removal. Moreover, a linear probe on the time-averaged state displacement, fitted to the teacher's rate change only on strengths up to the released ones, predicts the held-out stronger perturbations with $R^2 = 0.98$. At full inhibitory removal it predicts 12.15 Hz against the teacher's 13.83 Hz, whereas the model's own readout predicts 0.46 Hz. The control probe on input-count changes reaches only $R^2 = 0.44$ on the same held-out strengths. This indicates that the deficit lies in the mapping from strongly displaced states to spikes, which was fitted on histories whose firing rarely exceeds 10.5 Hz, the 99th percentile of ordinary rates, and not in a missing internal response. The probe measures linearly decodable information under supervision by the teacher; whether recalibrating the readout on high-rate histories would recover the response remains to be tested.

8 Population simulation

8.1 Throughput as a function of connectivity

The population benchmark advances 1,048,576 heterogeneous neurons at native 1 ms resolution. One real-time second therefore requires 1.048576×10^9 neuron-steps. The measured step includes event-derived firing, delayed contact routing, branch accumulation, mutable efficacies, recurrent state, and pinned-host telemetry for 1% of the neurons.

As expected from the per-contact routing cost (Section 3), throughput decreases as incoming synaptic connectivity increases (Figure 10). The million-neuron measurement uses 362 incoming contacts per neuron and advances at 1.177×10^9 neuron-steps/s. At 4,000 contacts, a closed-loop 65,536-neuron population advances at a median 2.215×10^8 neuron-steps/s, equivalent to 3.38 times real time. A routing-density sweep at the same population spans 100 to 10,000 contacts while holding source activity, external input width, and telemetry constant. Its 10,000-contact endpoint remains 2.17 times real time.

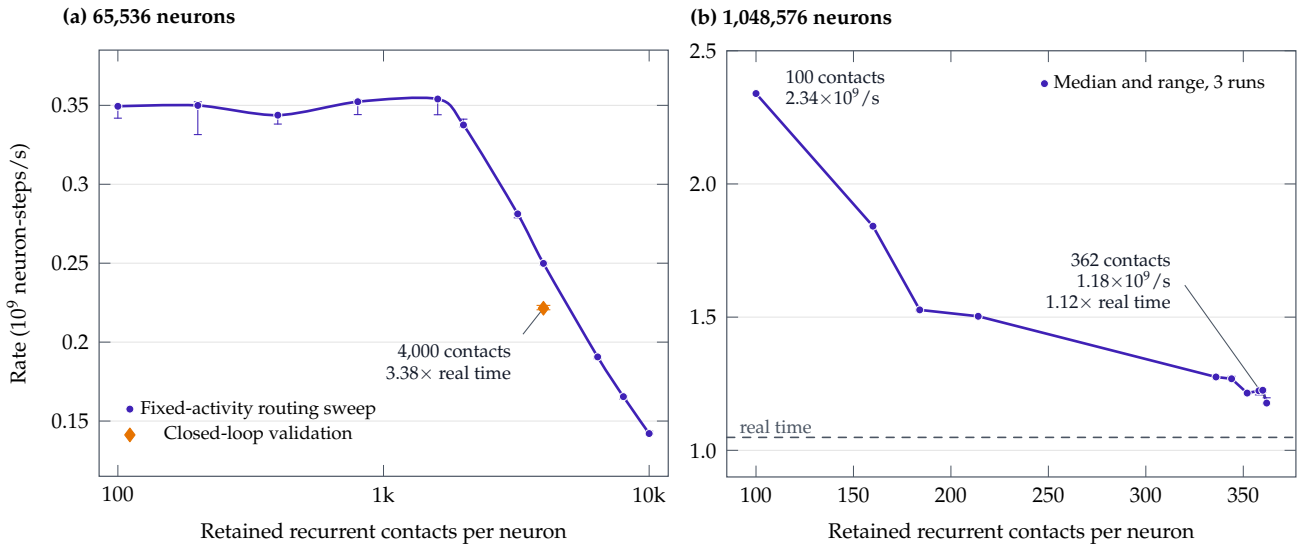


Figure 10: Throughput as a function of incoming connectivity. **(a)** A population of 65,536 neurons. Each marker is the median of three p90 rates, and vertical ranges span three independently realized graphs. The line connects measurements with source events replayed at morphology- and role-conditioned target rates, isolating routing density from changes in activity. The orange diamond validates the 4,000-contact point with threshold-derived closed-loop firing. **(b)** A population of 1,048,576 neurons with threshold-derived firing. Each marker is the median p90 rate of three runs in which all three reached the target firing rate, and vertical ranges span the three runs. The dashed line marks real time, 1.048576×10^9 neuron-steps/s. In both panels, every timed step includes the threshold scan, delayed routing, branch accumulation, an independently mutable efficacy for every contact, recurrent state, and telemetry for 1% of neurons.

The million-neuron measurement reports connectivity at a fixed population scale. The 65,536-neuron measurements extend the evidence to thousands of recurrent contacts and separate closed-loop execution from routing-density scaling. Appendix A.17 defines both workloads and reports the complete sweep.

8.2 AxoBench traces from recurrent population contexts

Recurrent feedback produces a distinct input distribution because event timing, excitation and inhibition, morphology, and graph structure jointly change branch load and temporal correlation. To sample that distribution, a thresholded AxoSim network generates branch-resolved contact histories, which are then replayed through matching detailed teachers. The recurrent learned system supplies the inputs, and five excitatory morphologies together with one inhibitory morphology supply detailed voltage and spike labels.

The `axobench-population` release contains 50,000 native-millisecond traces from 100 independently generated contexts. Eighty contexts provide 40,000 training traces, and 20 disjoint contexts provide 10,000 validation traces. Both splits preserve an 80/20 excitatory–inhibitory ratio. Validation recomputes split membership and checks manifest hashes, identifiers, array shapes, finite values, labels, and generator provenance. Paired perturbations reside in the separate `axobench-interventions` dataset, preserving the base population distribution.

Table 7: Composition of `axobench-population`. Complete recurrent contexts define the train–validation split.

Split and role	Traces	Contexts	Mean firing rate (Hz)
Train, excitatory	32,000	80	5.62
Train, inhibitory	8,000	80	6.32
Validation, excitatory	8,000	20	5.71
Validation, inhibitory	2,000	20	6.29
Total	50,000	100	–

8.3 Trace fidelity and recurrent validity are distinct

A plausible global firing rate can coexist with incorrect morphology-conditioned rates, temporal correlations, or late activity drift. A paired comparison isolates the effect of recurrent context by training one model on population-context examples and another on the original single-neuron distribution. On three unseen 4,000-neuron connectomes, population-context training reduces morphology discrepancy by 51.1%, temporal discrepancy by 7.2%, and stationarity discrepancy by 55.1%. It satisfies all three criteria fixed before evaluation, while the ordinary-input baseline misses the morphology criterion.

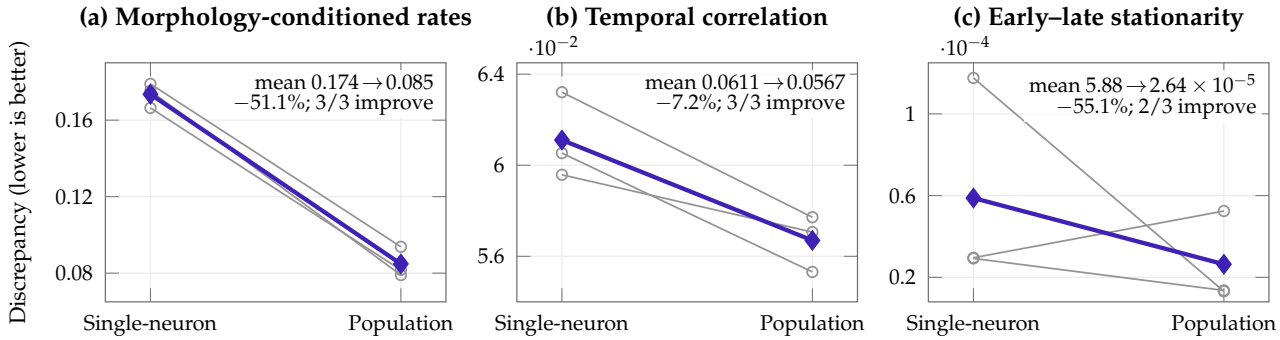


Figure 11: Development-corpus population-context training on three unseen 4,000-neuron connectomes. Thin lines pair training distributions on the same connectome; the purple diamond line connects cross-connectome means. Morphology measures mean squared log error in morphology-conditioned firing rates, temporal discrepancy measures lag-correlation error, and stationarity measures early–late log-ratio error. Annotations report the mean change and the number of improving connectomes. Lower is better throughout.

Training on all 50,000 traces improves held-out Voltage SERA from 299.99 to 285.84 mV^2, Dynamics SERA from 501.67 to 483.41 $\mathrm{mV}^2/\mathrm{ms}^2$, exact-spike F1 from 0.136 to 0.163, and Mean F1 from 0.331 to 0.340. Yet the connected model misses the morphology-conditioned criterion on all three unseen connectomes. Consequently, improved trace-level metrics do not imply recurrent validity. Connected measurements determine whether the gain transfers to recurrent deployment.

9 Inference-time adaptation

9.1 Structural and synaptic transfer

Five leave-one-morphology-out comparisons measure specialization of a small structural adapter to an excluded morphology. Each source uses the other four morphologies, whereas target adaptation and evaluation use disjoint stimulus schedules from the omitted morphology. Updating one 58-value row improves Mean F1 and Dynamics SERA on all five targets. Voltage SERA improves on three, while the other two change by only +0.27% and +0.37%. Full tuning of 38,257 parameters provides an upper reference for the attainable gain.

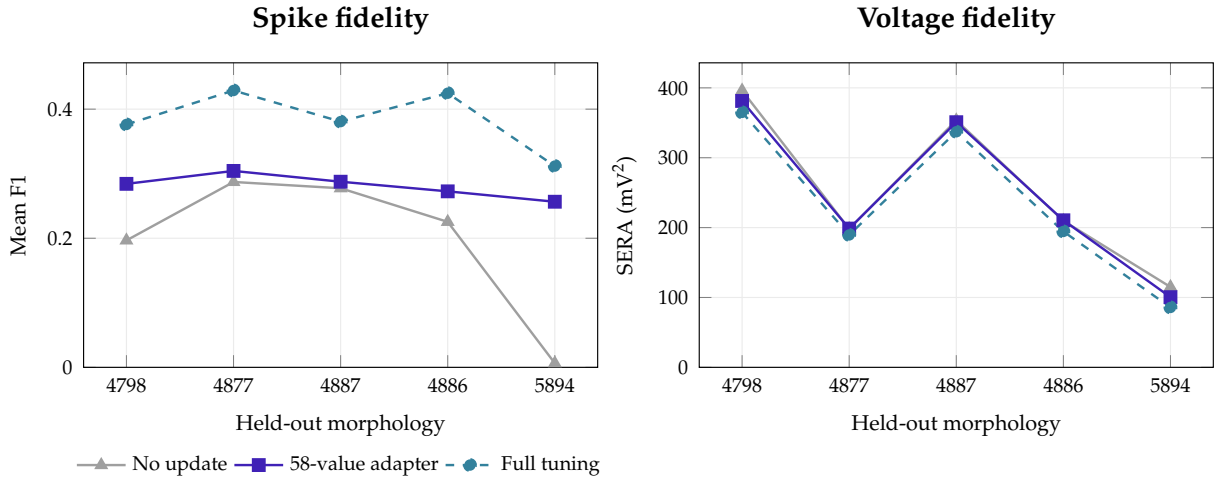


Figure 12: Adaptation to five separately excluded morphologies. Each line follows one adaptation alternative across the five held-out targets. The 58-value adapter improves spike and dynamics fidelity for every morphology; full tuning provides an upper reference.

Where morphology transfer targets an omitted class, synaptic transfer addresses a known perturbation. Two conditions of one Hay layer-5 pyramidal-cell teacher differ only by an excitatory-conductance scale of 0.925 versus 1.0, with all 1,278 physical input channels. In the 0.925-to-1.0 direction, contact adaptation reduces Voltage SERA by 4.76%, reduces Dynamics SERA by 0.31%, and raises Mean F1 by 0.044. The learned contact changes correlate 0.724 with the known conductance intervention. In the reverse direction, correlation rises to 0.834 but Voltage SERA worsens. In the forward direction, contact coordinates both track and predictively compensate for the controlled shift. The reverse direction demonstrates the insufficiency of parameter correlation alone for a biological interpretation.

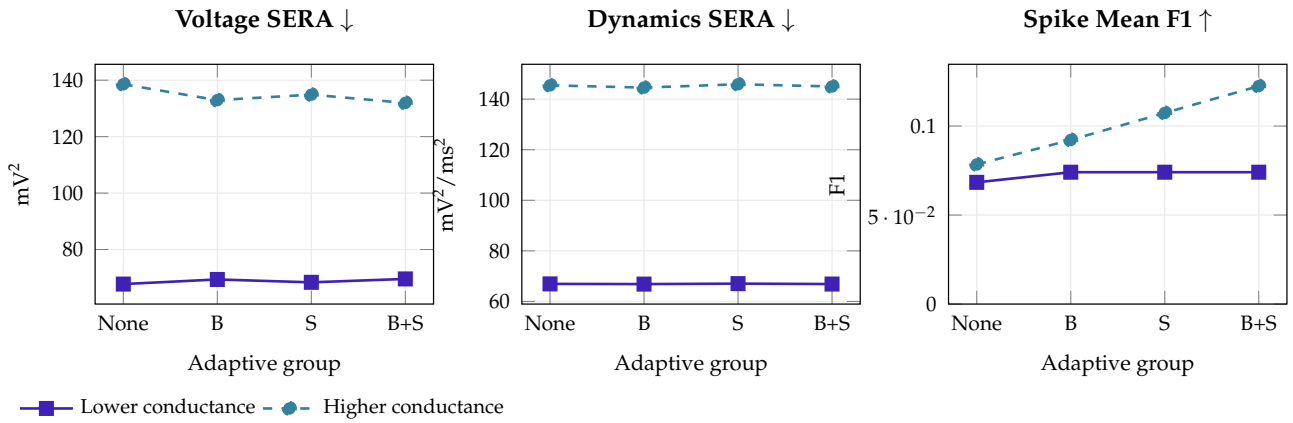


Figure 13: Transfer between matched conductance conditions of one detailed teacher. The 0.925-to-1.0 direction improves all three fidelity metrics. The reverse direction follows the imposed channel pattern more closely while increasing held-out Voltage SERA.

The role of each adaptive coordinate becomes concrete in these transfer comparisons. Updating a morphology row applies one class-shared correction, whereas synaptic updates change individual contact efficacies. Predictive improvement after a controlled shift demonstrates their practical utility. Cellular identification requires consistent predictive success across interventions.

9.2 Specialization to recurrent input distributions

Population-context inputs differ from the ordinary AxoBench distribution even when the neuron target is unchanged. Four AxoSim models and the released Branch-ELM family are adapted for 30,000 recurrent-domain presentations using the same context partition (Table 9) and architecture-specific schedules fixed on development data. Performance is then measured on 20 untouched target contexts.

Every AxoSim model improves target Mean F1. Mamba Medium reaches 0.412 after adaptation, compared with 0.136 for the strongest Branch-ELM target result. The other AxoSim models reach 0.364–0.402 and preserve the same large event-recovery separation. Target-domain results appear in Table 8; the full metric set is given in Appendix A.18.

Table 8: Spike fidelity after specialization to recurrent population inputs. The improvement is relative to the same model before target-domain adaptation.

Model	Target Mean F1	Improvement
AxoSim–GRU Small	0.379	+0.066
AxoSim–GRU Medium	0.402	+0.073
AxoSim–Mamba Small	0.364	+0.064
AxoSim–Mamba Medium	**0.412**	+0.094
Branch-ELM M100	0.136	+0.055

Corpus training targets a generally deployable recurrent model and therefore requires connected validation. Recurrent-domain adaptation instead measures whether exposed coordinates improve behavior in a specified target domain. The same data source supports both analyses under these distinct evaluation contracts.

9.3 Adaptation across a temporal task shift

Spiking Heidelberg Digits (SHD) supplies an external temporal distribution shift [7]. Development data compare seven optimizer–learning-rate–duration schedules and every nonempty combination of behavior, morphology, and synaptic groups. Morphology-plus-synapse adaptation, a learning rate of 2×10^{-3}, and 40 updates are fixed on one source model and reused for five independently trained sources. Each run uses 160 adaptation examples, 80 development examples, and 800 final examples, all disjoint.

Across six independently trained source models (Appendix A.5), target accuracy rises from 64.50% to 70.96%. The mean gain is 6.46 ± 1.20 percentage points, where the uncertainty is the sample standard deviation across six source models, and every target comparison improves. The source classifier used to choose the adaptation schedule reaches 91.88% before the temporal shift, localizing the lower target value to the shifted task.

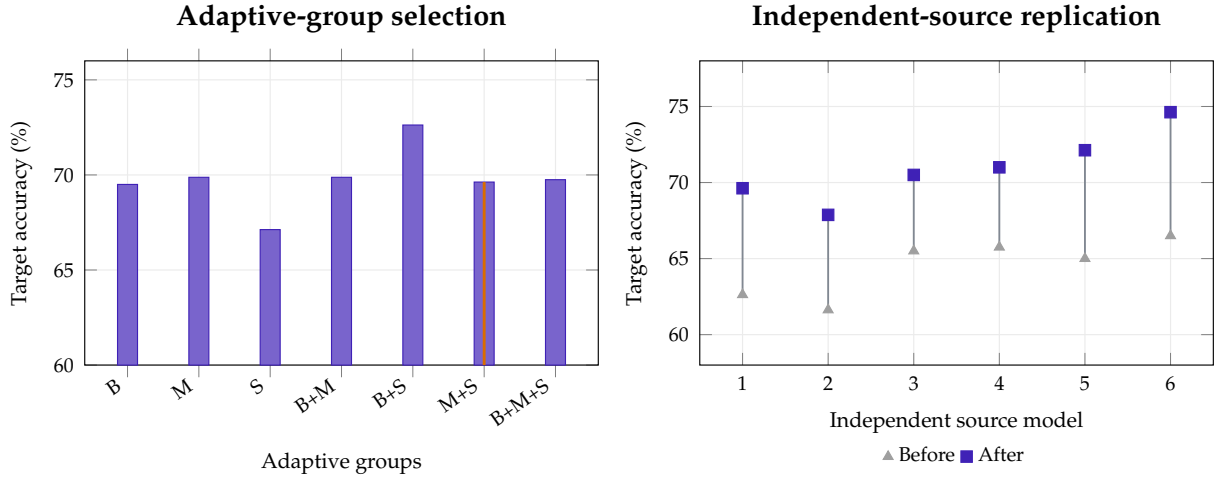


Figure 14: SHD accuracy before and after inference-time adaptation. One source model fixes the adaptive parameter groups and update schedule; five independently trained models reuse it. Target accuracy improves in all six comparisons.

Accuracy gains are non-monotonic in contact count. At 700 contacts, two independently trained source models improve from 65.13% to 72.13%, a gain of 7.00 ± 1.06 points, while their mean source validation accuracy is 92.56%. The complete contact-density analysis appears in Appendix A.19. The external task therefore provides replicated evidence of useful within-system adaptation. Comparison with separate test-time-adaptation algorithms remains future work.

9.4 Population size, connectivity, and update time

The runtime contract measures one complete 256-step adaptation update (Appendix A.2). Each replay includes fresh pinned-host input transfer, exact sparse contact aggregation, forward execution, loss, full backward propagation, gradient clipping, and one capturable-Adam update in BF16. Real-time adaptation requires all seven changing replays to finish within 256 ms.

At 2,000 synaptic contacts per neuron, all seven updates remain below the deadline for 20,000 neurons, with a slowest update of 252.17 ms. The next measured population, 21,000 neurons, exceeds the deadline. The tradeoff between synaptic contacts and speed (Section 3) also governs adaptation: doubling connectivity to 4,000 contacts reduces the real-time population to 10,000 neurons, whose slowest

update takes 250.50 ms. Behavior, morphology, and synaptic parameters are optimized together, and every synaptic contact has an independently trainable efficacy.

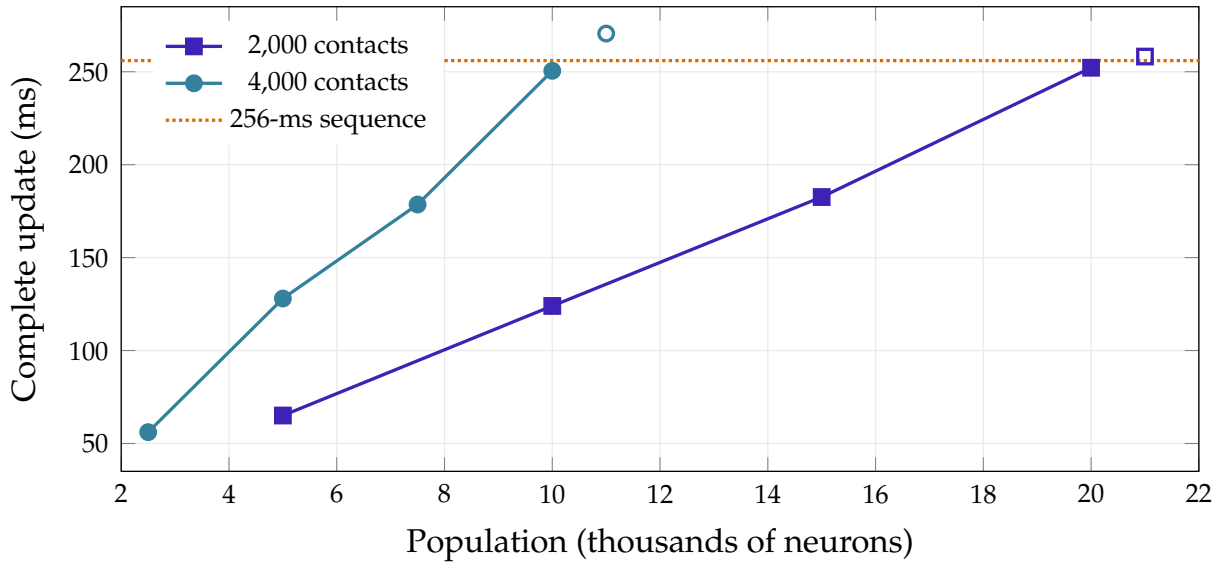


Figure 15: Complete full-BPTT update time for densely connected populations. Markers show the slowest of seven changing updates at each population size; filled markers below the dotted 256-ms boundary satisfy the real-time criterion, while open markers show the first measured misses. Each update includes fresh pinned-host input, exact sparse aggregation, forward execution, loss, full backward propagation, gradient clipping, and one Adam step with all adaptive parameter groups enabled.

Repeated fixed-shape updates therefore support full-sequence learning at the reported population and connectivity. Section 11 discusses sustained long-horizon online learning as the next research direction.

10 Relation to learned and mechanistic simulators

10.1 Learned and mechanistic neuron models

Temporal neural networks can approximate the millisecond input–output map of a morphologically detailed cortical neuron and amortize repeated numerical integration of the teacher [3]. Spieler et al. introduced the Expressive Leaky Memory neuron as a compact alternative in which dendritic integration and slow memory reproduce detailed-neuron responses with relatively few parameters [21]. AxoSim uses released Branch-ELM models as its principal learned reference and the Hay layer-5 pyramidal-cell family as a principal mechanistic teacher [10].

Arbor and CoreNEURON instead accelerate networks of morphologically detailed neurons while retaining compartments, membrane mechanisms, and their mechanistic interpretation [1, 14]. Jaxley makes detailed simulation differentiable with respect to explicit morphological, synaptic, and ion-channel parameters [8]. These systems remain appropriate for studying the effect of a named conductance or compartment. AxoSim amortizes detailed teacher simulations by fitting each learned model once to generated responses and then reusing it for population realizations and gradient-based specialization within the learned domain.

10.2 Simulation abstractions and population execution

Brian 2 separates model equations from generated execution code, allowing a scientific description to target efficient backends [22]. BrainPy similarly combines dynamical systems, transformations, simulation, training, and analysis in one programming model [26]. The same separation is important for learned neurons. AxoSim exposes ordinary PyTorch modules for model construction and differentiation, then maps the compact population model to fused kernels while preserving contact identity, delay semantics, mutable efficacy, and parameter ownership.

Population execution also changes what must be measured. Cortical synapse counts vary by species, region, layer, and cell type, with relevant values in the thousands across reported preparations [20]. Accordingly, AxoSim attaches each population throughput measurement to a specific number of incoming synaptic contacts. Connected validation then measures morphology-conditioned responses, temporal correlations, and stationarity as a distinct level of recurrent evidence.

10.3 Metrics for learned neuron simulation

AxoBench defines the core evaluation set used here, consisting of tolerance-dependent spike F1, Voltage SERA, and subthreshold error [27]; earlier AxoBench versions used Dynamics SERA in place of the subthreshold error. SERA integrates conditional squared error over teacher relevance and follows the imbalanced-regression construction of Ribeiro and Moniz [19]. Spike F1 is reported across explicit timing tolerances, complementing established spike-train distances [24, 25].

Each quantity measures a different aspect of the response. Spike F1 measures event recovery, Voltage SERA measures errors in relevant parts of the somatic trajectory, and the subthreshold error measures the voltage between action potentials. A model can therefore receive different rankings on the three metrics. For recurrent use, morphology-conditioned activity, temporal correlation, and stationarity diagnostics extend the metric set to population scale.

11 Discussion and future work

11.1 Recurrent use changes the unit of validation

Learned simulation creates a hierarchy of validation levels. Held-out traces measure voltage and spike prediction under externally specified input. Connected evaluation then measures behavior after those predictions alter the next input distribution, while adaptation differentiates the temporal response to a complete supplied history. Each level therefore requires its own evidence.

The distinction is visible in the population-data comparison. A controlled context corpus improves morphology, temporal, and stationarity behavior on unseen connectomes. Training on all 50,000 traces improves every held-out trace metric under the fixed training schedule, yet the connected model misses the morphology criterion. Small conditional errors can move recurrent state into a different input regime, and the graph then determines how those errors propagate. Consequently, a learned simulator should be validated at the level where its predictions become interventions.

These observations motivate an iterative division of labor. Detailed models generate teacher responses and controlled perturbations. Learned models amortize those responses across many connectomes or optimization steps. Connected diagnostics then identify distribution shifts and trigger new teacher simulations before the next deployment regime.

11.2 Edge-associated state dominates population scale

Parameter count of the shared neuron model is an incomplete predictor of population cost. At one million neurons, routing and branch accumulation grow from 59.7% to 78.8% of the component-median sum as incoming connectivity increases from 100 to 362 contacts. The fixed-activity routing sweep at 65,536 neurons then shows a broad plateau through 1,600 contacts, followed by a decline from 0.338 billion neuron-steps/s at 2,000 contacts to 0.142 billion at 10,000. Inference rate therefore depends jointly on population size, connectivity, activity, telemetry, and hardware.

Adaptation makes this relationship sharper because every retained edge also owns differentiable state. Doubling the synaptic contacts from 2,000 to 4,000 halves the measured real-time full-BPTT population from 20,000 to 10,000 neurons (Section 9). Compact shared weights can be reused across the population, whereas edge-associated routing, storage, and gradients scale with the realized graph. Reports of learned population simulators should therefore attach neuron count to contact count and update mode.

11.3 Learned coordinates require predictive interventions

Explicit parameter ownership permits optimization at neuron, structural-class and contact scales. Morphology rows improve transfer to five omitted structures, and synaptic values can follow a known conductance pattern. The class and neuron coefficients enter additively, so biological attribution requires predictive validation beyond ownership. In the reverse conductance transfer, parameter correlation is higher even though Voltage SERA worsens. A learned coordinate gains biological relevance when its change predicts the response to a controlled intervention.

The matched intervention comparison supplies direct output evidence through lower teacher-response error and teacher-aligned firing changes. Internal state geometry complements this predictive test. Its association with teacher-response distance rises from 0.211 in the branch representation to 0.446 after temporal integration, while the input perturbation reaches 0.471. The geometry consequently describes both the imposed input structure and its transformation through temporal memory. Together, output prediction, external response distances and controlled transfer relate learned coordinates to observable neuronal responses.

11.4 Future work

Population adaptation currently uses full backpropagation through time on supplied histories. Local learning rules, such as eligibility propagation [2] or spike-timing-dependent plasticity [4], would update the same parameter banks from signals available forward in time. They would remove the activation memory of full backpropagation through time and could adapt a population in closed loop on the spikes it generates itself, which the current gradient does not reach.

The AxoBench metrics are also unsaturated. Replicated training should examine depth, width, residual conditioning, objective balance, and compute while improving spike Mean F1, Voltage SERA, and Dynamics SERA together. In the measured frontier, the largest temporal core approaches the spike leader while smaller models lead the remaining metrics. Controlled, replicated trajectories can determine how depth, width, and compute produce that tradeoff.

Adaptation should extend from bounded transfer episodes to sustained long-horizon online learning. Such a study must measure cumulative task performance, memory growth, numerical stability, and drift over many updates. A matched-simulator comparison should fix morphology, stimulus, connectivity, precision, and output requirements across AxoSim and a detailed engine. Together, these

analyses would strengthen both the scientific validity boundary and the practical interpretation of the runtime frontier.

12 Conclusion

AxoSim combines learned single-neuron fidelity with causal sparse population execution and differentiation through neuron- and contact-associated state. These capabilities are usually evaluated separately. Its strongest accuracy result is spike timing, accompanied by lower Voltage and Dynamics SERA than released Branch-ELM references. Paired synaptic interventions also yield lower teacher-response error across all compared model sizes. At population scale, a complete million-neuron step runs in real time at 362 contacts per neuron. A 65,536-neuron population remains 3.38 times faster than real time at 4,000 contacts, while complete full-BPTT updates remain real time for 20,000 neurons with 2,000 contacts per neuron.

The results also yield a methodological conclusion. Held-out trace gains and connected-population validity form two distinct levels of evidence because recurrent predictions change their own future inputs. AxoSim makes that distinction measurable by preserving contact semantics, exposing the ownership of adaptive state, and attaching each result to its measurement conditions.

Code and documentation. Installation, model and population interfaces, inference-time adaptation, worked examples, and the API reference are available at https://axo.axym.org.

References

[1] Nora Abi Akar, Ben Cumming, Vasileios Karakasis, Anne Küsters, Wouter Klijn, Alexander Peyser, and Stuart Yates. Arbor—a morphologically-detailed neural network simulation library for contemporary high-performance computing architectures. *arXiv preprint arXiv:1901.07454*, 2019. URL https://arxiv.org/abs/1901.07454.

[2] Guillaume Bellec, Franz Scherr, Anand Subramoney, Elias Hajek, Darjan Salaj, Robert Legenstein, and Wolfgang Maass. A solution to the learning dilemma for recurrent networks of spiking neurons. *Nature Communications*, 11:3625, 2020. doi: 10.1038/s41467-020-17236-y.

[3] David Beniaguev, Idan Segev, and Michael London. Single cortical neurons as deep artificial neural networks. *Neuron*, 109(17):2727–2739.e3, 2021. doi: 10.1016/j.neuron.2021.07.002.

[4] Guo-qiang Bi and Mu-ming Poo. Synaptic modifications in cultured hippocampal neurons: Dependence on spike timing, synaptic strength, and postsynaptic cell type. *Journal of Neuroscience*, 18(24):10464–10472, 1998.

[5] Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*, pages 1724–1734, 2014. doi: 10.3115/v1/D14-1179. URL https://aclanthology.org/D14-1179/.

[6] Mark M. Churchland, John P. Cunningham, Matthew T. Kaufman, Justin D. Foster, Paul Nuyujukian, Stephen I. Ryu, and Krishna V. Shenoy. Neural population dynamics during reaching. *Nature*, 487(7405):51–56, 2012. doi: 10.1038/nature11129.

[7] Benjamin Cramer, Yannik Stradmann, Johannes Schemmel, and Friedemann Zenke. The heidelberg spiking data sets for the systematic evaluation of spiking neural networks. *IEEE Transactions on Neural Networks and Learning Systems*, 33(7):2744–2757, 2022. doi: 10.1109/TNNLS.2020.3044364.

[8] Michael Deistler, Kyra L. Kadhim, Matthijs Pals, Jonas Beck, Ziwei Huang, Manuel Gloeckler, Janne K. Lappalainen, Cornelius Schröder, Philipp Berens, Pedro J. Gonçalves, and Jakob H. Macke. Jaxley: Differentiable simulation enables large-scale training of detailed biophysical models of neural dynamics. *Nature Methods*, 22(12):2649–2657, 2025. doi: 10.1038/s41592-025-02895-w.

[9] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*, 2023. URL https://arxiv.org/abs/2312.00752.

[10] Etay Hay, Sean Hill, Felix Schürmann, Henry Markram, and Idan Segev. Models of neocortical layer 5b pyramidal cells capturing a wide range of dendritic and perisomatic active properties. *PLOS Computational Biology*, 7(7):e1002107, 2011. doi: 10.1371/journal.pcbi.1002107.

[11] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. *arXiv preprint arXiv:1512.03385*, 2015. URL https://arxiv.org/abs/1512.03385.

[12] Michael L. Hines and Nicholas T. Carnevale. Translating network models to parallel hardware in NEURON. *Journal of Neuroscience Methods*, 169:425–455, 2008.

[13] Nikolaus Kriegeskorte, Marieke Mur, and Peter A. Bandettini. Representational similarity analysis—connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2:4, 2008. doi: 10.3389/neuro.06.004.2008.

[14] Pramod Kumbhar, Michael Hines, Jeremy Fouriaux, Aleksandr Ovcharenko, James King, Fabien Delalondre, and Felix Schürmann. CoreNEURON: An optimized compute engine for the NEURON simulator. *Frontiers in Neuroinformatics*, 13, 2019. doi: 10.3389/fninf.2019.00063.

[15] Leland McInnes, John Healy, and James Melville. UMAP: Uniform manifold approximation and projection for dimension reduction. *Journal of Open Source Software*, 3(29):861, 2018. doi: 10.21105/joss.00861.

[16] Razvan Pascanu, Tomas Mikolov, and Yoshua Bengio. On the difficulty of training recurrent neural networks. In *Proceedings of the 30th International Conference on Machine Learning*, volume 28, pages 1310–1318, 2013. URL https://proceedings.mlr.press/v28/pascanu13.html.

[17] Michael Pearce, Elana Simon, Michael Byun, and Daniel Balsam. Finding the tree of life in Evo 2. Goodfire Research, 2025. URL https://www.goodfire.com/research/phylogeny-manifold. Published August 27, 2025.

[18] PyTorch Contributors. CUDA semantics: CUDA graphs. PyTorch 2.7 documentation, 2026. URL https://docs.pytorch.org/docs/2.7/notes/cuda.html. Accessed October 2, 2026.

[19] Rita P. Ribeiro and Nuno Moniz. Imbalanced regression and extreme value prediction. *Machine Learning*, 109(9–10):1803–1835, 2020. doi: 10.1007/s10994-020-05900-9.

[20] Almut Schüz and Günther Palm. Density of neurons and synapses in the cerebral cortex of the mouse. *Journal of Comparative Neurology*, 286(4):442–455, 1989. doi: 10.1002/cne.902860404.

[21] Aaron Spieler, Nasim Rahaman, Georg Martius, Bernhard Schölkopf, and Anna Levina. The expressive leaky memory neuron: An efficient and expressive phenomenological neuron model can solve long-horizon tasks. *arXiv preprint arXiv:2306.16922*, 2023. URL https://arxiv.org/abs/2306.16922.

[22] Marcel Stimberg, Romain Brette, and Dan F. M. Goodman. Brian 2, an intuitive and efficient neural simulator. *eLife*, 8:e47314, 2019. doi: 10.7554/eLife.47314.

[23] David Sussillo and Omri Barak. Opening the black box: Low-dimensional dynamics in high-dimensional recurrent neural networks. *Neural Computation*, 25(3):626–649, 2013. doi: 10.1162/NECO_a_00409.

[24] M. C. W. van Rossum. A novel spike distance. *Neural Computation*, 13(4):751–763, 2001. doi: 10.1162/089976601300014321.

[25] Jonathan D. Victor and Keith P. Purpura. Metric-space analysis of spike trains: Theory, algorithms and application. *Network: Computation in Neural Systems*, 8(2):127–164, 1997. doi: 10.1088/0954-898X/8/2/003.

[26] Chaoming Wang, Tianqiu Zhang, Xiaoyu Chen, Sichao He, Shangyang Li, and Si Wu. BrainPy, a flexible, integrative, efficient, and extensible framework for general-purpose brain dynamics programming. *eLife*, 12:e86365, 2023. doi: 10.7554/eLife.86365.

[27] Davide Wiest. AxoBench: Evaluating learned single-neuron surrogates as mechanistic simulators. Manuscript in preparation; public benchmark repository, 2026. URL https://github.com/Axym-Labs/axobench.

A Measurement contracts and supplementary evidence

A.1 Evaluation partitions

Table 9 gives the disjoint fitting, development, and final-evaluation units used by the principal comparisons.

Table 9: Data used for parameter fitting, schedule selection, and final evaluation.

Comparison	Fitting or adaptation	Development	Final evaluation
Ordinary AxoBench	pretraining corpus	120 balanced traces	next 120: 24 threshold, 96 scored
Intervention response	fixed models	50 baseline calibration pairs	350 paired histories, 175 shards
Population specialization	64 recurrent contexts	16 recurrent contexts	20 untouched contexts
SHD adaptation	160 examples per source	80 examples on one source	800 examples on six sources
Connected populations	training traces	development connectomes	three paired unseen connectomes

A.2 Runtime measurement contracts

A.3 Model dimensions and optimization trajectories

The four full models use the dimensions below, which were fixed before the final pretraining run.

Pretraining uses AdamW with batch size 32, initial learning rate 6×10^{-4}, cosine decay, and weight decay 10^{-4}. The objective combines scaled somatic-voltage MSE with weight 0.75, spike binary cross-entropy with weight 0.25, a direct SERA surrogate with weight 0.01, a differentiable F1 tolerance-

Table 10: Work included in each reported runtime quantity.

Quantity	Horizon and statistic	Included work	Excluded work
Isolated inference	500 steps, repeated median	model forward	data loading
Forward and backward	batch 32, 500 steps, 20 repeats	zero-grad, forward, loss, backward	loader, optimizer
Population inference	P4 advances, end-to-end p90	firing, routing, state, efficacy, 1% telemetry	data generation, initialization
Population adaptation	256 steps, seven replays	transfer, sparse input, forward, full BPTT, clip, Adam	graph capture

Table 11: Temporal-core dimensions of the retained full models.

Model	Model width	State width	Layers	Head width
GRU Small	32	8	2	32
GRU Medium	128	32	2	128
Mamba Small	32	8	2	32
Mamba Medium	176	44	4	88

curve term over 0–5 steps with weight 0.05, and near-spike voltage error within five steps with weight 0.002. The SERA surrogate evaluates 101 relevance thresholds.

Pretrained states at 24, 32, and 40 million presentations are compared on 120 development traces. GRU Small is retained at 32 million presentations, whereas the other three profiles are retained at 40 million. A separate set of 120 morphology-balanced traces supplies the final evaluation block.

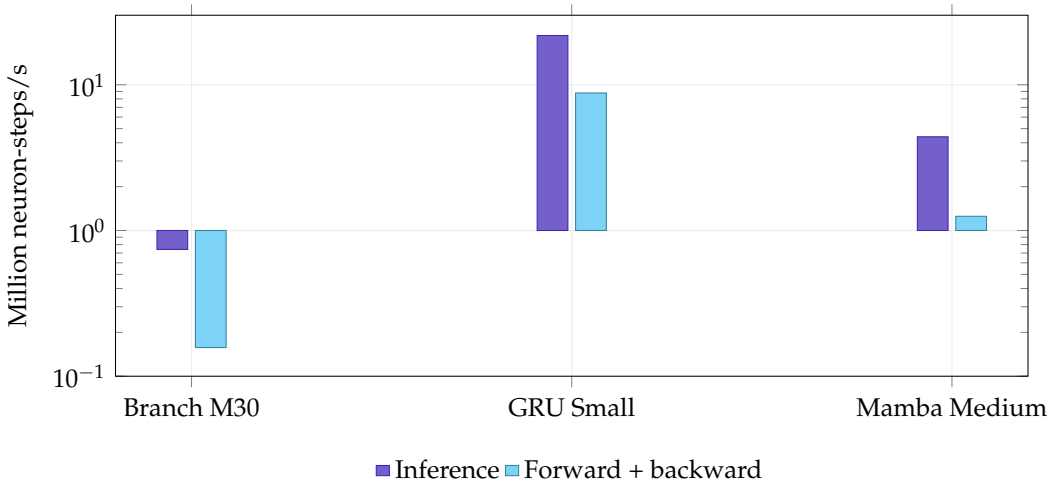


Figure 16: Paired inference and forward-plus-backward throughput for one representative of each architecture family. Each rate is reconstructed from the stored median latency, batch size, and 500-step window; loader and optimizer work are excluded.

A.4 AxoSim–Lite pretraining

The population benchmarks of Section 8 and the SHD and runtime experiments of Section 9 use one pretrained AxoSim–Lite checkpoint with token width 24, state width 8 and a four-step forecast. It is pretrained on the ordinary AxoBench training split, covering the five evaluation morphologies, for 300,000 morphology-balanced presentations at batch size 32, starting from slices of a wider four-step GRU surrogate. The objective combines the detailed-teacher spike, voltage, F1-curve and SERA terms with distillation from an internal four-step Mamba forecaster, which is not one of the reported AxoSim–Mamba models. Optimization uses AdamW with learning rate 6×10^{-4}, cosine decay, weight decay 10^{-4} and gradient clipping at 1, and the checkpoint is the end of this fixed bud-

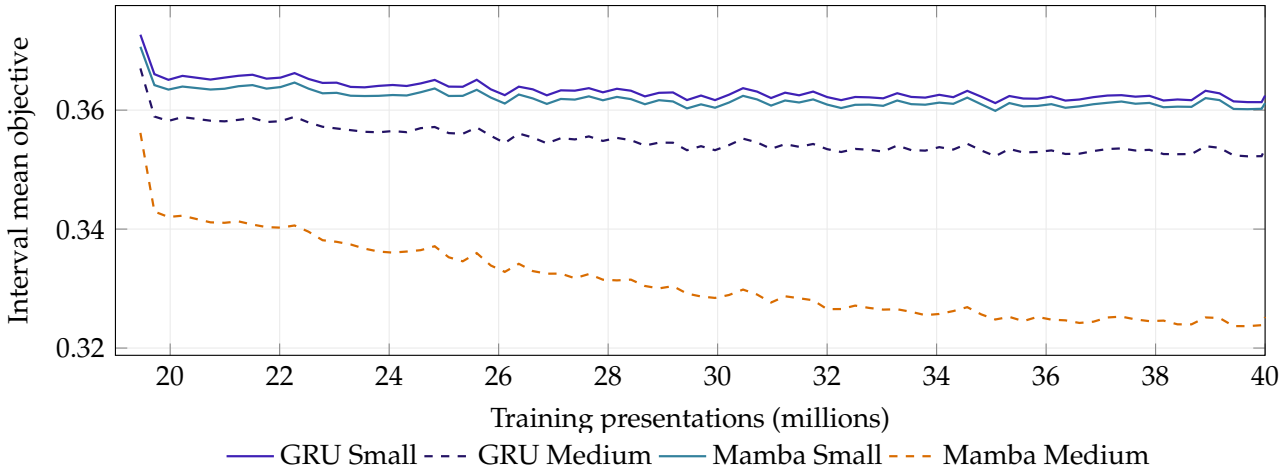


Figure 17: Interval mean optimization objective for the four stable models after the common training run resumed near 19.5 million presentations. The interval view distinguishes late plateauing and batch variation from numerical divergence.

get. On 400 validation traces, it reaches Voltage SERA 225.83 mV^2, Dynamics SERA 360.03 mV^2/ms^2 and Mean F1 0.354; these traces differ from the confirmation block of Table 4. The adaptation timing runs use these weights with synthetic sparse inputs and a squared-output loss, since they measure the duration of an update rather than its effect.

A.5 SHD source models

Each SHD source model keeps the pretrained Lite neuron frozen and trains a population of 12 neurons with 426 contacts each, its coefficient banks and a one-hidden-layer readout with cross-entropy, selected on source-validation accuracy. The six source models differ in their random seeds and source splits.

A.6 Matched component ablations

The component comparison retrains ten two-layer Mamba configurations with 91,451 total parameters, 300,000 presentations and 9,375 updates per model, using seed 20260826 and the same five-morphology sampling schedule. Ablations disable one component while retaining its allocated parameters; the near-identity branch-saturation control freezes the affected transform parameters. The evaluation contains 120 traces, with 30 used for threshold calibration and 90 for scoring after the first 500 ms. Five hundred paired trace-bootstrap resamples estimate evaluation uncertainty for this common training seed. Table 12 reports the five components discussed in Section 2.

These records use Voltage Root-SERA in mV and the trapezoidal average of F1 across 0–5 ms. They therefore differ from the current Voltage SERA in mV^2 and arithmetic Mean F1. For every table column, positive values favor the retained component. Paired intervals for F1 at 5 ms include zero for morphology embedding and branch saturation, whereas their Voltage Root-SERA and trapezoidal-F1 intervals favor retention.

Table 12: Component-removal penalties after matched retraining. Entries give the point estimate and paired 95% interval. Voltage penalties are ablated minus full; F1 losses are full minus ablated.

Removed component	Voltage Root-SERA (mV)	Trapezoidal F1	F1 @ 5 ms
Route conditioning	.423 [.370, .477]	.087 [.072, .102]	.081 [.063, .099]
Morphology embedding	.207 [.162, .256]	.017 [.003, .032]	.008 [−.009, .025]
Local temporal refinement	.125 [.090, .163]	.016 [.003, .028]	.023 [.008, .038]
Branch saturation	.105 [.090, .123]	.009 [.002, .017]	.010 [−.001, .020]
Spike–voltage coupling	.091 [.066, .116]	.016 [.005, .027]	.017 [.004, .031]

The matched configuration uses width 58, state width 23, two temporal layers and 118 branches, with shared synaptic gain, bilinear interaction and a soma gate. Its readout differs from the separate heads used by the reported full models. A further 88,858-parameter configuration jointly removes these three interactions and uses separate heads; at batch 32 it is 13.0% faster, with a 0.021 mV increase in Voltage Root-SERA and a 0.003 increase in trapezoidal F1. This combined change informs the streamlined parameterization; the individual components' necessity remains conditional on the readout and training configuration.

A.7 Compact adaptive coefficients

For token width D, recurrent width H, horizon P and output width O, the compact recurrence uses $Q = 2D + 3H + 2PO$ coefficients. Table 13 defines their packed groups. With $\widetilde{u} = (u + o_u) \odot (1 + g_u)$, its exact transition and readout are

$$\begin{aligned} d &= \sigma(\lambda + \delta_d), \\ p &= \tanh(W_u \widetilde{u} + W_h h + b_h + o_h) \odot (1 + g_h), \\ h' &= d \odot h + (1 - d) \odot p, \\ \widehat{y} &= (W_o h' + b_o + W_s \widetilde{u}) \odot (1 + g_o) + o_o. \end{aligned}$$

Here λ is the shared decay logit; $\widehat{y}$ is reshaped into P native predictions with O coordinates each.

Table 13: The seven deployed coefficient groups. The measured compact model uses $D = 24, H = 8, P = 4, O = 2$, giving $Q = 88$.

Coefficient	Count	Symbol	Role
Token offset	D	o_u	input offset
Token gain	D	g_u	input sensitivity
Decay shift	H	δ_d	memory timescale
Proposal gain	H	g_h	recurrent proposal amplitude
Proposal offset	H	o_h	recurrent proposal offset
Output gain	PO	g_o	spike/voltage readout scale
Output offset	PO	o_o	spike/voltage readout offset

For neuron i of structural class $c(i)$, the coefficient vector is $q_i = a_i + m_{c(i)}$, where a_i is the behavior bank and m_c is the class bank. This additive parameterization permits selective optimization at either scale. Adding a vector to one class row and subtracting it from all corresponding neuron rows leaves the response unchanged, so unique attribution requires an additional constraint. Contact parameters instead act through positive efficacies $\exp(s_{i,k})$ before input aggregation. These coordinates describe predictive response changes; biological interpretation rests on their performance under controlled transfer.

A.8 Contact and neuron memory

Let K be incoming contacts, B branch features, R delay-ring slots, H recurrent coordinates and Q adaptive coefficients. If b_s, b_a, b_c are the bytes used for an efficacy, activation and coefficient, respectively, the principal persistent banks require $Kb_s + RBb_a + Hb_a + Qb_c$ bytes per neuron, plus Qb_c per structural class. The measured ring has $R = 8, B = 118$; BF16 ring entries occupy 1,888 bytes per neuron and the $H = 8$ recurrent state occupies 16 bytes. This accounting excludes shared weights, forecast buffers, routing workspace, telemetry and allocator overhead.

Table 14: Bank-level memory costs for $K = 4{,}000, B = 118, R = 8, H = 8, Q = 88$. The coefficient examples use BF16; contact deployment uses int8, while the training row includes FP32 values, gradients and two Adam moments. These are bank costs, rather than total GPU allocation.

Bank	Bytes	Example
Deployed contact efficacies	K per neuron	4,000
Recurrent state	$2H$ per neuron	16
Delayed branch ring	$2RB$ per neuron	1,888
Neuron coefficients (BF16)	$2Q$ per neuron	176
Class coefficients (BF16)	$2Q$ per class	176
Contact values, gradients and Adam state	$16K$ per neuron	64,000

Deployed efficacies are represented as signed 8-bit deviations from a positive baseline with a shared scale, requiring K bytes per neuron. Floating-point adaptation uses a separate representation. With FP32 efficacies, gradients and two Adam moment buffers, contact-associated parameter and optimizer storage is $16K$ bytes per neuron, before temporal activations. Thus 2,000 contacts contribute 32,000 bytes per neuron in this training accounting. Sparse event accumulation removes the dense $N \times T \times K$ input history, while full temporal BPTT retains the activations needed across the supplied T-step sequence.

A.9 Native-time population contracts

The compact forecast consumes exactly one completed $P = 4$ input patch. The first four native samples are a startup prefix with no valid forecast and no emitted recurrent events. Thereafter each output has its own native timestamp. Every synaptic contact is validated to have delay at least 4 ms, and the delay ring covers both the minimum delay and forecast horizon. Neuron phases are assigned independently of morphology and source role within stratified tiles.

The stateful population interface maintains neuron state, the delayed queue and mutable contact efficacies across calls. Supplied-history tensor execution provides the differentiated sequence, whereas the procedural phase/tile-major implementation supplies the measured large-population inference path. General explicit-contact graphs and the structured procedural topology share contact semantics but have distinct implementations and runtime measurements. Floating-point efficacy in the general interface is separate from the quantized efficacy bank used by the deployed timing profile.

A.10 Complete-update capture parity

The eager/captured SHD comparison performs 40 updates from identical source parameters and optimizer state on the same ordered batches. Both paths use capturable Adam, the same learning rate, cross-entropy objective, adaptive parameters and clipping rule. Capture warms up on a side stream, records the complete update, and restores the model and optimizer states before replay. The maxi-

mum final parameter discrepancy is zero. Eager execution takes 6.404 s, replay 1.836 s and separate capture preparation 0.737 s. These times establish parity and replay acceleration for this workload; the dense-contact full-BPTT timings use their own measurement contract.

A.11 Matched intervention-response comparison

The comparison selects the first two pairs from each of 200 dataset shards, using pair ordinal alone. Five interventions and five morphologies define 25 strata, with eight shards per stratum. The first shard of every stratum supplies calibration, giving 50 pairs; the remaining seven supply 350 scoring pairs in 175 shard clusters. Selection is independent of model predictions and preserves equal condition and morphology counts.

Each model's single prevalence-matched spike threshold is calibrated from baseline predictions and baseline teacher spikes only, then frozen for both members of every scoring pair. The four full AxoSim and five released Branch-ELM models retain their selected weights. Each six-second history is evaluated from a fresh initial state at 1-ms resolution, with the first 500 ms excluded. There are 70 scoring pairs and 385 scored neuron-seconds per intervention condition.

The NeuronIO voltage coordinate is $\widehat{v} = 0.1(v + 67.7)$, so predictions are converted once by division by 0.1 and subtraction of 67.7 mV. Fidelity uses raw, unclipped teacher soma voltage. Training targets are clipped at -55 mV; their consistency check is $\text{target}/0.1 - 67.7 = \min(v_{\text{raw}}, -55)$. This establishes the saved target relation while preserving the raw-teacher metric policy.

For each pair, let $\Delta v = v^{\text{intervention}} - v^{\text{baseline}}$. Response SERA applies the conditional squared-error integral to $\Delta\widehat{v} - \Delta v$, with relevance equal to the maximum of baseline-voltage, intervention-voltage and teacher-response-tail relevance. Counts, squared errors and cross-products are pooled before computing the summary metrics. The zero-response control sets $\Delta\widehat{v} = 0$. Centered Pearson correlation describes response-shape association, while the least-squares gain $\sum \Delta\widehat{v}\,\Delta v / \sum(\Delta v)^2$ describes the projection of the predicted change onto the teacher change. The uncentered response cosine is retained separately in the numerical records.

One thousand paired bootstrap draws resample shards within each of the 25 strata, retaining the two dependent traces within each selected shard. Every draw recomputes pooled sufficient statistics for all nine models. The primary family contains all twenty AxoSim–Branch-ELM response differences. A maximum absolute standardized bootstrap statistic provides simultaneous 95% intervals across this family, with critical value 2.478. For example, GRU Small minus Branch-ELM M1 is $-30.62\,\text{mV}^2$, with simultaneous interval $[-32.47, -28.78]$. All twenty intervals are below zero. These intervals quantify shard-sampling uncertainty for the fixed models and morphologies. Table 15 gives individual model intervals, distinct from these paired family intervals. Figure 7 shows descriptive pooled firing-rate changes, without interpolation between the unordered intervention categories.

Table 15: Absolute fidelity and response diagnostics on the 350 scoring intervention pairs. Response SERA is in mV^2, with individual 95% bootstrap intervals; the zero-response control gives 298.27. Pearson and gain use the same raw-teacher voltage changes.

Model	Response SERA	95% interval	Mean F1	Pearson	Gain
AxoSim–GRU Small	255.55	[248.44, 262.29]	.439	.487	.276
AxoSim–GRU Medium	256.30	[249.19, 263.30]	.470	.485	.269
AxoSim–Mamba Small	256.31	[249.31, 263.11]	.416	.483	.271
AxoSim–Mamba Medium	258.78	[251.47, 265.76]	.491	.485	.303
Branch-ELM M1	286.17	[278.71, 293.70]	.070	.336	.183
Branch-ELM M10	287.85	[280.38, 295.27]	.062	.331	.179
Branch-ELM M30	287.67	[280.15, 295.10]	.076	.333	.183
Branch-ELM M50	287.90	[280.35, 295.34]	.073	.333	.186
Branch-ELM M100	287.92	[280.41, 295.35]	.075	.332	.186

The teacher-input mapping assigns synaptic channels to dendritic sites using normalized arbor length. Site silencing removes a declared band of input channels, so its physical interpretation is a perturbation of synaptic drive at mapped sites. Event removal and sign-selective removal alter the corresponding input events; jitter alters their times. The response results therefore concern these five input interventions, with the same teacher mapping used for baseline and intervention.

A.12 Intervention dose series

Each dose series starts from the 70 scoring pairs of one released condition, namely the first two pairs of the seven scoring shards of each morphology. The released baseline input and baseline teacher response are retained. The intervention seed of each pair is reconstructed from the generator, $20260611 + 1009\,s + i$ for shard index s and pair index i, and regenerating the released strength reproduces every released intervention input exactly. Because event removal compares one uniform draw per event with the removal probability, the events removed at a lower strength are a subset of those removed at a higher strength. Site silencing removes all events in a contiguous band of sites starting at the first site.

The 910 additional teacher histories use CPU NEURON with the released settings: Hay biophysics, deterministic AMPA/NMDA and GABA synapses with unscaled conductances, an initial voltage of -76 mV, a 0.025-ms integration step, a 1-ms synaptic delay and arclength-normalized routing. Resimulating one released baseline and one released intervention history for each morphology reproduces the released voltages with a maximum absolute difference of 0.0 mV.

All models, checkpoints, spike thresholds, the 500-ms exclusion and the response statistics are those of the matched comparison. The released TCN checkpoints receive binary inputs and their somatic output is interpreted as $V + 67.7$ mV, following the original training code; on the original NeuronIO test traces, this interface reproduces the published spike AUC of 0.99 and an unaligned clipped-voltage RMSE of 0.7 mV. Each TCN threshold is prevalence-matched on the same 50 calibration baselines, and the checkpoint with the lowest Response SERA pooled over the three conditions at the released strengths is shown, which is the rule that selects Branch-ELM M1.

Intervals resample the seven shards of each morphology with replacement and retain both pairs of a selected shard, giving 1,000 draws. They describe uncertainty over histories for fixed models and morphologies. The temporal state of Mamba Medium is the output of its temporal core after the first 500 ms. Each of its channels is divided by its standard deviation over the baseline histories, and

the displacement of one pair is the RMS over time and channels of the intervention-minus-baseline standardized state.

The ordinary reference comprises the 12,500 histories of the validation split of the training generator. For the probes, each pair and strength contributes one example whose target is the teacher's firing-rate change. The state features are the time average of the intervention-minus-baseline temporal state after 500 ms, and the control features are the changes in excitatory and inhibitory event counts per second. Ridge regressions with penalty 10 on standardized features are fitted on the 560 examples at or below the released strength of each condition and evaluated on the 560 examples above it.

A.13 Intervention-response geometry

The intervention analysis uses the final AxoSim–Mamba Medium model and all 12,500 paired traces in `axobench-interventions`. The five intervention families are event dropout, excitatory dropout, inhibitory dropout, site silencing, and temporal jitter, crossed with five morphologies and eight dataset shards per combination. Baseline state is subtracted from intervention state for every pair. The first 500 ms are excluded, and the remaining 5,500 ms are averaged into eleven consecutive windows.

The signal column in Figure 2 uses the 500-ms window with the highest teacher spike count among one deterministically selected pair from each of the 200 shards. This trace exposes measured input, state, and output signals in the architecture diagram. It is excluded from every reported aggregate statistic. The input display uses 10-ms event counts, while the temporal representation, voltage, and spike probability are sampled at the native 1-ms resolution. The displayed probability is the sigmoid of each individual logit. PC 1 is computed after standardizing each temporal feature within this window.

Panel (a) groups each shard deterministically into disjoint mini-cohorts of seven or eight pairs, producing 1,575 points. Each point concatenates the mean 176-dimensional temporal-state displacement in the eleven windows. Features are standardized before fitting a three-dimensional cosine-distance UMAP with 200 neighbors, minimum distance 1.2, spread 1.5, default repulsion strength 1.0, and random seed 20260930. Coordinates are mean-centered for display. These unsupervised display parameters were selected from ten retained candidates; 15-neighbor trustworthiness is 0.9716, compared with 0.9722 for 30 neighbors and minimum distance 0.12. No edges between intervention families are added, and quantitative response comparisons use the original representations.

Panel (b) uses the eleven retained temporal windows of the same 1,575 disjoint cohorts. Each cohort mean is weighted by its seven or eight constituent pairs when standardizing features, fitting PCA, averaging condition paths, and estimating flow. The first three components explain 59.6%, 27.0%, and 6.9% of this weighted cohort-mean variance, rather than individual-trace variance. The 15,750 observed next-window cohort transitions are assigned to a $10 \times 10 \times 10$ grid spanning the 3rd–97th percentile range; out-of-range starts enter the corresponding boundary bins. Each arrow begins at the weighted mean observed start and represents the weighted mean displacement of that bin. All 202 bins with at least eight cohort transitions are displayed, accounting for 15,267 transitions. A common arrow-length multiplier of 0.28 preserves their relative lengths and directions while reducing overlap. This display-only resampling leaves the original-space geodesic, direction, and output-response metrics unchanged.

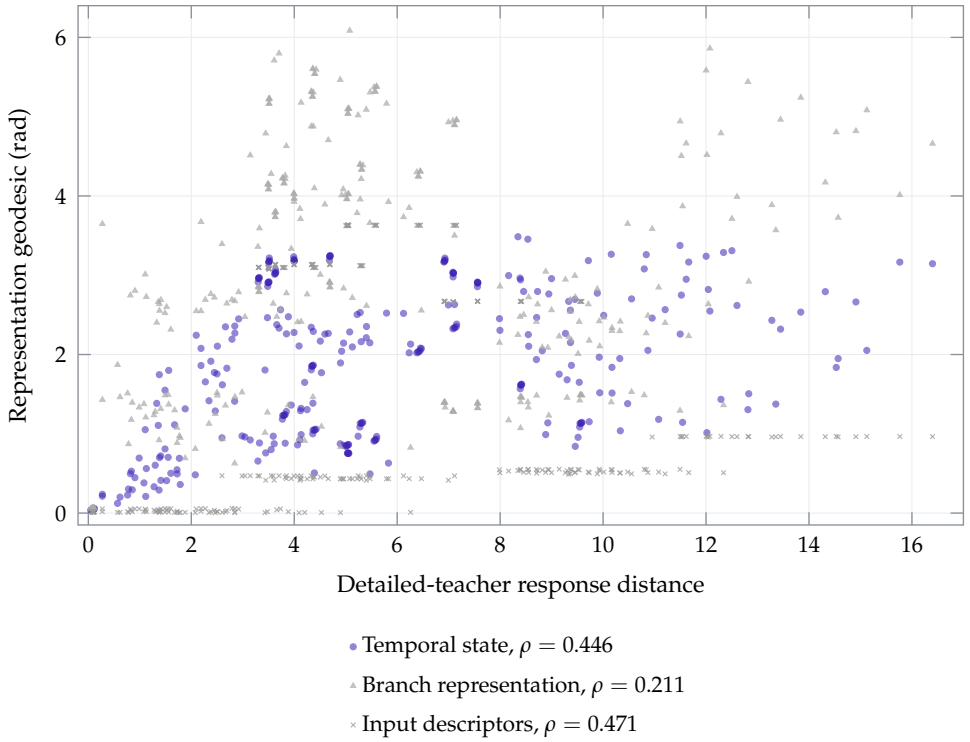


Figure 18: Association between teacher-response and representation distances across 25 intervention–morphology groups, or 300 group pairs. Geodesics use each original standardized representation; teacher distances use standardized voltage and spike-response signatures. Spearman correlations are 0.446 for temporal state, 0.211 for branch features and 0.471 for input descriptors. The input control relates the observed response organization to information in the imposed perturbation.

The external-ruler comparison averages the eight shards within each of the 25 groups defined by intervention and morphology. In each representation, feature-standardized angular distances form the smallest connected symmetric nearest-neighbor graph. The required neighbor counts are five for branch state, six for temporal state, and five for the input descriptor. Shortest-path distances are then compared by Spearman correlation with Euclidean distance between standardized detailed-teacher signatures. Each teacher signature contains eleven mean voltage changes and eleven spike-count changes. The 95% intervals come from 1,000 resamples of the eight shards within every group; each resample reconstructs the group representations and graph. The direction-consistency margin subtracts the mean cosine similarity across interventions within a morphology from the mean similarity across morphologies within an intervention.

The direction-consistency margin is 0.563, with a 95% interval of [0.559, 0.566]. Teacher-distance Spearman correlations and their intervals are 0.446 [0.443, 0.451] for temporal state, 0.211 [0.167, 0.213] for branch features and 0.471 [0.443, 0.498] for input descriptors.

A.14 Figure 1 values and axis mappings

Table 16 gives the values represented in Figure 1; `reproducibility/central_figure_values.py` derives every plotted radius from the frozen evidence. Inference throughput uses the absolute mapping $1 - \exp(-r/10^7)$, with r in neuron-steps/s. History throughput uses $1 - \exp(-r_h/10^6)$, with r_h in supplied input milliseconds per second; this counts input duration, not sequential implementation steps. The three fidelity axes are plotted untransformed and clipped to $[0, 1]$. They are the dataset-

relative scores of Section 6.2: voltage SERA skill as a fraction of the clipped-teacher skill, subthreshold RMSE skill with cell-type action-potential windows, and Mean F1 over 0–5 ms, each on 0–6 Hz traces of the model's own training data, reweighted to AxoBench's firing-rate distribution. These mappings are fixed independently of the compared values. The Branch–ELM panel shows M1, M30, and M100, covering the smallest released model, a middle size, and the largest released model, which also has the highest Mean F1. The TCN panel shows the smallest checkpoint, which also has the highest Mean F1, and the largest. All released models use the same evaluation and timing contracts; the others are listed below rather than overlaid on their almost coincident polygons.

Table 16: Raw values used by Figure 1. Inference rates are millions of neuron-steps/s and history rates are millions of input-ms/s. Fidelity values are rate-matched on each model's own training data with 95% bootstrap intervals (Table 6).

Variant	Inference	Voltage fidelity	Subthreshold fidelity	Mean F1	History
CoreNEURON reference	0.00085	1	1	1	0.000031
Branch-ELM M1	0.788	0.753	0.552	0.206	0.0289
Branch-ELM M10	0.717	0.877	0.739	0.312	0.0312
Branch-ELM M30	0.742	0.901	0.773	0.421	0.0296
Branch-ELM M50	0.735	0.908	0.779	0.444	0.0319
Branch-ELM M100	0.712	0.909	0.776	0.447	0.0315
TCN 7_128_153	2.955	0.878	0.714	0.374	0.736
TCN 7_292_169	2.754	0.871	0.727	0.375	1.635
TCN 8_224_217	4.706	0.869	0.721	0.379	1.949
TCN 9_256_241	3.817	0.852	0.669	0.379	1.706
AxoSim-Mamba Medium	4.396	0.908	0.718	0.598	2.053
AxoSim-GRU Medium	16.810	0.937	0.705	0.607	2.011
AxoSim-GRU Small	21.786	0.936	0.690	0.572	4.216

CoreNEURON supplies the target trajectories, so it defines the top of every fidelity axis. Its inference rate comes from a different hardware and workload record and appears as an absolute measurement rather than the denominator of a cross-hardware speedup. Learned-model rates use the shared 500-step inference contract described in Section 5.

History throughput measures one supplied 2,000-ms history on an RTX 5090, using the same signed input-event tensor for all systems. Each of the 1,278 contacts receives Bernoulli events with probability 0.005 per millisecond. Learned models use FP32, ten warmups and forty continuous CUDA-event timings of the complete forward and readouts; loading and input transfer are excluded. All eight learned AxoSim and Branch–ELM rates come from one session; an earlier session measured GRU Small 8% slower and the other models within 5%. The TCN session re-timed Branch–ELM M30 within 1% of that earlier session. CoreNEURON uses the Hay cell with deterministic synapses, 0.025-ms internal integration and its preserved 6.3 °C temperature. Its synchronous wall-time measurement includes the CoreNEURON handoff, device transfer, voltage recording and spike extraction, while cell construction and initialization are excluded. Three complete replays are timed, with biological state reinitialized before each. Source hashes, repeated latencies and readout checks accompany the numerical evidence. The released TCN checkpoints run through a PyTorch port of their Keras weights under the same inference and history contracts, with cuDNN's default convolution precision. Their released code processes a long history in overlapping fixed-length windows; Figure 1 uses a single causal pass over the 2,000-ms history instead, which reproduces the windowed outputs to within 2×10^{-6} and is 5–8 times faster. These different measurement boundaries do not establish a kernel-only simulator speedup.

The learned-model comparison concerns fitted quality and measured execution rates. AxoSim and the released Branch-ELM models received different upstream optimization budgets, so these measurements do not establish relative training efficiency.

A.15 Pareto-arrow selection and FLOP contract

The orange arrow in each panel of Figure 5 follows one fixed rule. Within the Branch–ELM group and the combined AxoSim group, points dominated on both the displayed efficiency and fidelity axes are discarded. The remaining coordinates are min–max normalized within their group after orienting both axes so that larger values are better. The selected point is the nondominated candidate with the smallest Euclidean distance to the ideal corner ((1,1)), with model identifier as a deterministic tie-breaker. The precomputed endpoint table in `data/tikz/pareto-arrows.csv` decouples selection from figure layout.

Forward-pass FLOPs count the operations recognized by `torch.profiler` for one batch-one, 500-step sequence with 1,278 input features, together with one-dimensional convolutions, which the profiler does not count. Each convolution contributes two operations per multiply-accumulate, that is, twice the number of output elements times the input channels per group times the kernel width. This term covers the TCN layers and the convolutions of AxoSim–Mamba. The same measurement is applied to every learned model. Parameter counts include trainable parameters only. Neither quantity includes data loading or optimizer operations.

A.16 Evaluation on the NeuronIO cell

The cross-dataset comparison scores all 1,536 traces of the 12 NeuronIO test files [3] with the configuration of the ordinary comparison: the first 500 ms are ignored, one prevalence-matched spike threshold is calibrated on the first 20% of traces, Voltage SERA uses the raw somatic voltage and Mean F1 averages tolerances of 0–5 ms. The teacher spike labels are those of NeuronIO. Branch-ELM and the TCNs receive the signed and binary encodings of their training code, and AxoSim is scored once with each of its five morphology identities, none of which is the NeuronIO cell. As an interface check on 377 traces of the first three test files with at most 24 teacher spikes, the TCN 7_128_153 and Branch-ELM M1 reach spike AUC 0.992 and 0.951 against the published-pipeline values of 0.991 and 0.944. The reference prediction for the voltage skill sets every scored voltage to the mean clipped teacher voltage of the calibration traces; it yields SERA_0 of 181.36 mV^2 on NeuronIO and 356.46 mV^2 on the AxoBench confirmation block. Its constant spike score ties with the calibrated threshold, so every step counts as a predicted spike, and the resulting Mean F1 values $\mathrm{F1}_0$ of 0.002 and 0.007 equal the values implied by the spike prevalence of each dataset. They define the spike skill $(\mathrm{F1} - \mathrm{F1}_0)/(1 - \mathrm{F1}_0)$.

The dataset-relative comparison (Table 6) scores each family on its own training data: AxoSim on the AxoBench confirmation block and Branch-ELM and the TCNs on the NeuronIO test traces, with the calibration traces above excluded. Only traces with teacher rates below 6 Hz, which both datasets cover, are scored: 73 of 96 AxoBench traces and 1,227 of 1,229 NeuronIO traces. NeuronIO traces are reweighted so that the rate bins 0–1, 1–3 and 3–6 Hz carry AxoBench's shares of 0.178, 0.452 and 0.370. Voltage SERA uses IRon relevance per spike, and the voltage skill is divided by the skill of the teacher voltage clipped at -55 mV, the best prediction without action potentials; after reweighting, this ceiling is 0.531 on AxoBench and 0.680 on NeuronIO. The subthreshold skill is $1 - \mathrm{RMSE}/\mathrm{RMSE}_0$ on voltage clipped at -55 mV outside teacher and predicted action-potential windows, with a per-trace-mean reference RMSE_0; each window starts 2 ms before a spike and ends, per cell type, where the teacher's mean spike waveform first falls below 1 mV/ms in slope. Sums over traces are pooled with the rate

weights. The 95% intervals are percentile bootstraps over 1,000 resamplings of scored traces, stratified by cell and rate bin.

A.17 Population timing and connectivity

Both population measurements use one RTX 5090 and BF16 compact-neuron execution at 1 ms resolution. Each timed step includes delayed routing that preserves source identity, mutable positive efficacies, recurrent state, and pinned-host voltage telemetry for 1% of neurons. The million-neuron measurement uses closed-loop spike discovery and 362 synaptic contacts per neuron. Its procedural graph partitions the neurons into tiles of 4,096; 90% of each neuron's outgoing contact channels target its own tile with delays of 4 or 5 ms, and the remaining channels target another tile with delays of 6 or 7 ms. Three target-activity replicates span 1.174–1.197 billion neuron-steps/s, with a median of 1.177 billion neuron-steps/s.

Figure 10 uses 65,536 neurons to measure denser graphs at 100, 200, 400, 800, 1,600, 2,000, 3,200, 4,000, 6,400, 8,000, and 10,000 retained recurrent contacts per neuron. A threshold scan remains in the timed step, after which source events are replayed at fixed morphology- and role-conditioned target rates. This replay holds activity constant while the measurement varies routing density. External input width remains fixed at 1,278 afferents, and every recurrent contact retains an independently mutable efficacy. Repeated source–target pairs are permitted, as multiple synapses may connect the same neuron pair.

Each contact count is measured on three independently realized graphs. The reported value is the median p90 step rate, with the full replicate range shown by the error bars. In this fixed-activity routing sweep, the 4,000-contact median is 249.87 million neuron-steps/s and the 10,000-contact median is 142.09 million neuron-steps/s, equivalent to 2.17 times real time. The curve connects measured medians and terminates at the 10,000-contact endpoint.

The orange 4,000-contact marker replaces source replay with threshold-derived closed-loop firing. Across three graph realizations, the measured-to-target rate ratio spans 1.021–1.026 with fixed thresholds throughout the measured window. The median p90 rate is 221.47 million neuron-steps/s, equivalent to 3.38 times real time for 65,536 neurons.

Quantitative mouse-neocortex estimates are on the order of thousands of contacts per neuron, with a reported average near 8,200 and substantial region and cell-type variation. Contacts also differ from counts of unique presynaptic partners. Population size, connectivity, and simulated time per wall clock second should therefore be reported together.

A 426-contact implementation used a stride-three bit-mask selector. Bit-mask reduction is valid only for power-of-two moduli; with a non-power-of-two fanout it collapses residue classes and does not realize the declared destinations. Correct modulo routing restores every class and yields more scattered target addresses. In the directly comparable records, median routing time rises from 1.579 to 1.965 ms. The increase is consistent with both the arithmetic remainder and the less-local target addresses; the timing record does not identify their individual contributions.

A.18 Population-input specialization details

Recurrent-domain specialization complements the ordinary-domain comparison. Section 6 gives the direct cross-family SERA result on ordinary AxoBench. The target-domain analysis instead measures how each model adapts its spike behavior to recurrently generated inputs.

Table 17: Target-domain metrics after 30,000 population-context presentations. SERA values use the same squared units as the ordinary AxoBench metrics.

Model	Mean F1	Voltage SERA	Dynamics SERA
AxoSim–GRU Small	0.379	48.10	20.24
AxoSim–GRU Medium	0.402	50.34	23.49
AxoSim–Mamba Small	0.364	47.89	21.34
AxoSim–Mamba Medium	0.412	49.23	24.23
Branch-ELM M50	0.132	36.87	16.36
Branch-ELM M100	0.136	36.93	16.36

A.19 SHD contact-density analysis

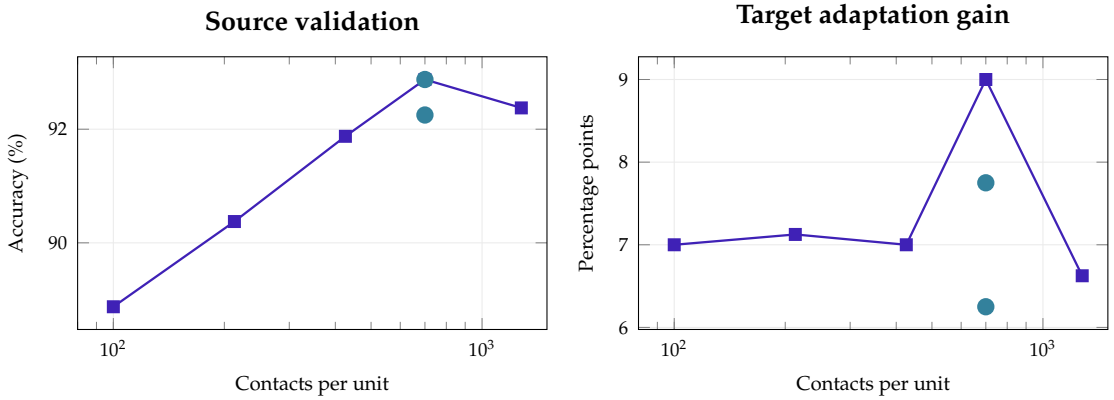


Figure 19: SHD adaptation as the number of synaptic contacts changes. The development result on one source is non-monotonic; two independently trained sources confirm the selected 700-contact setting.

A.20 Connected diagnostic definitions

Let $r_{s,m}$ denote the firing rate of excitatory morphology group m over either the full or tail window s, and let $r^\star_m$ denote its contract target. The morphology discrepancy is

$$D_{\text{morph}} = \frac{1}{2|\mathcal{M}|} \sum_{s\in\{\text{full,tail}\}} \sum_{m\in\mathcal{M}} \left[\log\left(\frac{r_{s,m}}{r^\star_m}\right)\right]^2.$$

For per-neuron input-count lag correlations $\rho_{s,\ell}$, evaluated at $\ell \in \{1, 4, 16, 64, 256\}$ ms, the temporal discrepancy is

$$D_{\text{temp}} = \frac{1}{2|\mathcal{L}|} \sum_{s} \sum_{\ell\in\mathcal{L}} \left(\frac{\rho_{s,\ell} - \rho^\star_\ell}{0.2}\right)^2.$$

Finally, let q range over global rate, excitatory rate, inhibitory rate, and input-count coefficient of variation. The stationarity discrepancy is

$$D_{\text{stat}} = \frac{1}{4} \sum_{q} \left[\log\left(\frac{q_{\text{tail}}}{q_{\text{full}}}\right)\right]^2.$$

Targets and normalizations are fixed before held-out testing. The three quantities remain separate throughout evaluation.

A.21 Conditional single-neuron examples

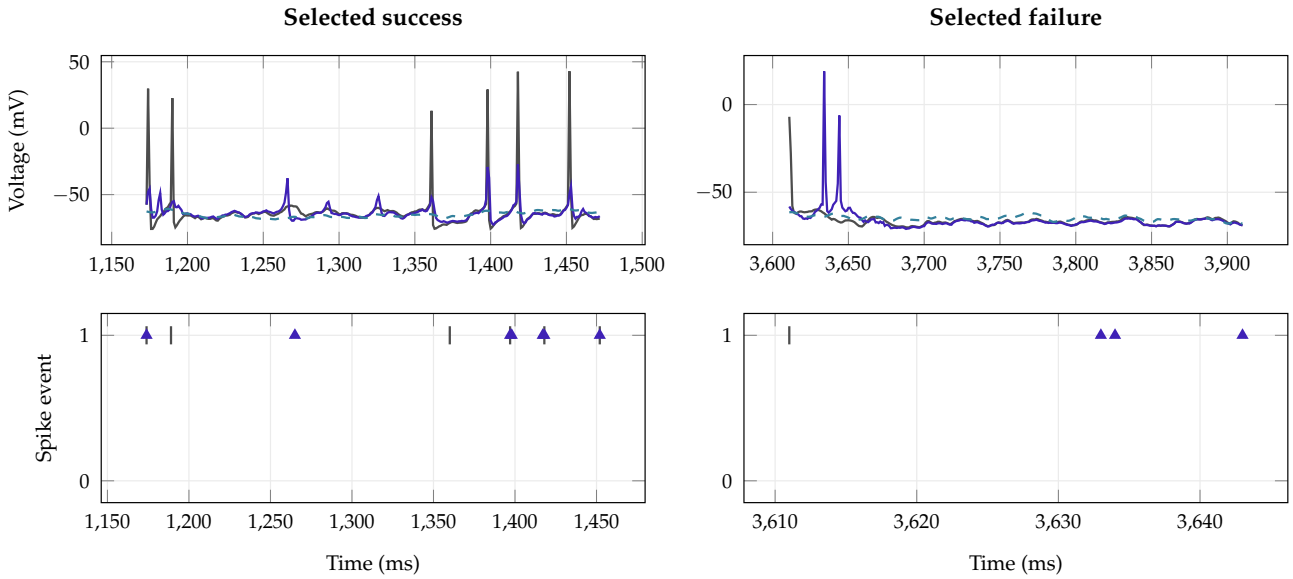


Figure 20: Balanced conditional examples from the final comparison block. The upper row is the spike-containing 300 ms window with the largest voltage-MSE advantage for the selected AxoSim–Mamba model over the strongest Branch-ELM spike baseline; the lower row is the largest disadvantage under the same rule. These selected examples expose one success and one failure and are not estimates of average performance. Black denotes the detailed teacher, solid purple AxoSim, and dashed cyan Branch–ELM.

A.22 Inference and full-BPTT contact memory

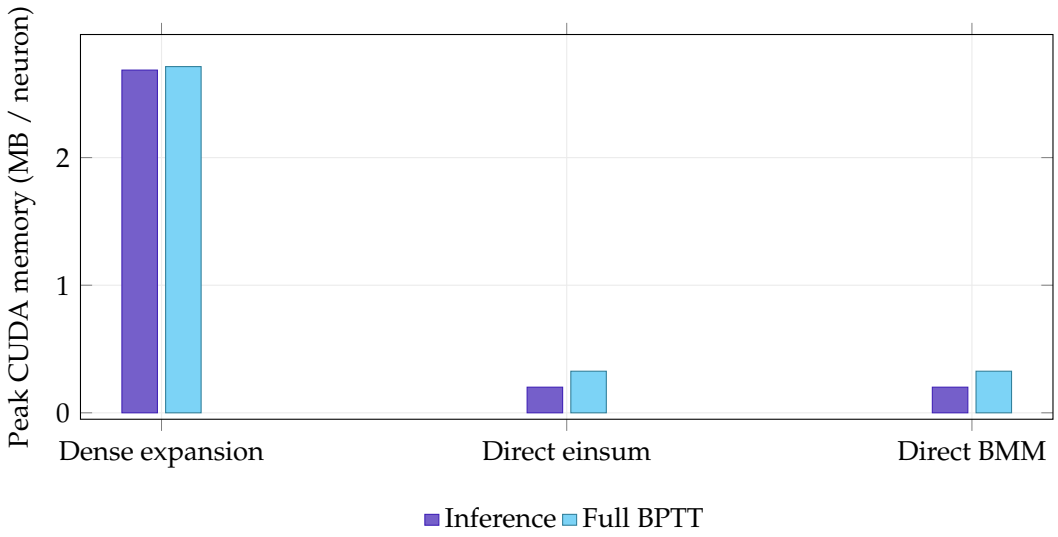


Figure 21: Peak CUDA memory per neuron for inference and full BPTT across dense branch expansion, direct einsum, and direct BMM contact-input implementations at 24 contacts and 256 timesteps.

Dense branch expansion uses approximately 2.686 MB per neuron for inference and 2.713 MB for full BPTT. Direct einsum and direct BMM each use approximately 0.201 MB for inference and 0.326 MB

for full BPTT. Normalized per-neuron memory permits comparison across the different executable population sizes; the source CSV also records raw populations and byte counts.

A.23 Adaptation timing boundary

The dense-contact benchmark optimizes behavior, morphology, and synaptic parameters together through 256 native-millisecond steps. At 2,000 synaptic contacts per neuron, 20,000 neurons pass all seven changing updates and 21,000 exceed the 256-ms deadline. At 4,000 contacts, 10,000 neurons pass and 11,000 exceed the deadline. The reported curve uses the slowest update at each population size. Timing includes fresh pinned-host input, exact sparse contact aggregation, transfer, forward execution, loss, full backward propagation, gradient clipping, and Adam; one-time graph capture is excluded.